

Strategy-Proofness Made Simpler ^{*}

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Abstract

There is evidence that many people do not report their types truthfully in strategy-proof mechanisms. One of the leading explanations is mechanism complexity. We propose a novel way of describing *any* strategy-proof mechanism with the aim to reduce the ensuing cognitive load on players and, hence, to increase truthful reporting. The description highlights that a player is guaranteed to obtain a most preferred feasible allocation based on her reported type and that feasibility depends only on the other players' reports. Optimality of truthful reporting should therefore become more "apparent." We experimentally test this prediction using the top trading cycles mechanism for matching applicants to positions. The proposed description increases truth-telling. This increase positively interacts with subject numeracy.

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JEL Classification: C72, C91, D44, D47, D91

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1 Introduction

Strategy-proof direct revelation mechanisms are hallmark achievements of market design. They are designed to allow (private or public) organizations to overcome the problem of allocating resources based on private information of customers, suppliers or other stakeholders, in general referred to as “players,” which those might be reluctant to reveal. This information typically involves preferences, costs, productivities or beliefs, in general referred to as “types.” Players are simply asked to report their type and the mechanism provides incentives to report truthfully. A player in a strategy-proof mechanism does not need to form beliefs about other players’ reports or types. She can simply report her type truthfully because she can do no better regardless of what the other players report or what their types are. That is, truthful reporting is a weakly dominant strategy. There is growing evidence, however, that players often “misreport” their types in strategy-proof mechanisms. For example, there is ample experimental laboratory evidence that many bidders overbid their value in second-price sealed-bid auctions (SPAs) with private values ([Kagel, Harstad and Levin, 1987](#); [Kagel and Levin, 1993](#); [Harstad, 2000](#); [Andreoni, Che and Kim, 2007](#)). Or, in student-school (or analogous) matching mechanisms based on deferred acceptance (DA, [Gale and Shapley \(1962\)](#)), top trading cycles (TTC, [Shapley and Scarf \(1974\)](#), originally proposed by Gale, see also [Abdulkadiroğlu and Sönmez \(2003\)](#)) or (random) serial dictatorship ((R)SD), participants frequently misreport their true ordinal preferences both in the lab ([Chen and Sönmez, 2006](#); [Pais and Pintér, 2008](#); [Li, 2017](#)) and in the field ([Gross, DeArmond and Denice, 2015](#); [Hassidim, Romm and Shorrer, 2021](#); [Chen and Pereyra, 2019](#); [Rees-Jones, 2018](#); [Rees-Jones and Skowronek, 2018](#); [Artemov, Che and He, 2023](#); [Shorrer and Sóvágó, 2023](#)).¹ Such misreporting has a potential to reduce efficiency or other intended properties of the resulting allocation or to introduce a bias or noise into inference about population distribution of types that might be useful for other purposes. It is therefore important to ask what the mechanism designer can do to improve the truth-telling rate.

Our point of departure is the hypothesis that misreporting is to a large extent caused by players’ cognitive limitations in combination with mechanism complexity. To discover that truthful reporting is dominant, a player needs to observe that reporting truthfully is a best response in any contingency defined as a possible collection of other players’ type reports and a realization of a tie-breaking device.² Some players might find this challenging because computing allocations resulting from various combinations of type reports and contingencies is difficult. Moreover, a large amount of contingencies typically needs to be considered. In addition, complexity also hampers an alternative approach of by-passing these computational challenges by identifying

¹[Hassidim et al. \(2021\)](#), [Artemov et al. \(2023\)](#) and [Shorrer and Sóvágó \(2023\)](#) establish this observation relying purely on observational data on reported rankings, whereas [Gross et al. \(2015\)](#), [Chen and Pereyra \(2019\)](#), [Rees-Jones \(2018\)](#) and [Rees-Jones and Skowronek \(2018\)](#) complement the observational data with a survey to elicit true ordinal preferences.

²For example, in a student-school matching mechanism, a contingency is a collection of reported rankings of the other students and school priority sequences. In a multi-unit Vickrey auction, a contingency is a collection of sets of bids of the other bidders and bid tie-breakers.

the strategy-proofness property directly from the mechanism description (as one would do in a proof).

With the aim of making it easier for players to realize that reporting truthfully is dominant, we propose a novel “pick the best feasible allocation,” or, in short, “pick-the-best,” description of *any* strategy-proof mechanism. This description does not alter the mechanism. In particular, to each player, the mechanism is described as follows: “In step 1, based on the reported types of the other players (and, possibly, realization of some tie-breakers), the mechanism determines which allocations you can feasibly get (your *feasible allocations*), i.e., it determines your *feasible set*. Your reported type plays no role in this step. In step 2, the mechanism selects for you an allocation from your feasible set that you prefer the most according to your reported type.”³ The player is also given access to the description of how the feasible set is constructed in step 1, but this information is de-emphasized.

To illustrate, consider two examples. In student-school matching, the feasible set of a student consists of a subset of schools into which the student can make it under some own report. This set is independent of the student’s report and *ex ante* unknown. Assuming that students have strict preferences over schools, the mechanism then interprets the student’s ranking as her true strict preference order and picks a feasible school that is highest up the student’s submitted ranking. In the Vickrey auction, the feasible set of a bidder consists of offers to buy various subsets of the items at prices that are exogenous and *ex ante* unknown to this bidder. The mechanism then interprets the bidder’s bids as her values (i.e., maximum willingness to pay) and picks a subset of items that maximizes the bidder’s surplus.

Now consider how the pick-the-best description helps mitigating the impact of complexity discussed above. In Step 1, the description maps the original space of (potentially abstract and high-dimensional) contingencies into a space of feasible sets of allocations which now replace contingencies as the states to consider. In Step 2, the interaction between a type report and a feasible set is simplified to picking a report-optimal allocation from the feasible set. There are several implications. First, since feasible sets and their interaction with the type report are arguably easier to understand and implement computationally than determining an allocation resulting from a type report and a contingency, a player is more likely to successfully utilize contingent thinking. Second, the space of feasible sets is coarser than the space of contingencies as a feasible set typically represents many contingencies, hence reducing computational complexity in this regard. Third, the description allows a simple direct identification of the strategy-proofness property: “If the mechanism always picks, according to my declared preferences, a best allocation from some set of allocations unaffected by my report, why not declare my preferences truthfully?”

We empirically identify the impact of the pick-the-best description on truth-telling rates

³Step 2 requires that one’s type must be a sufficient statistic for determining which allocation(s) are preferred the most in any possible feasible set. Without a restriction on what such feasible sets might look like for a given mechanism, this requirement is in general equivalent to the assumption of “private values”.

by examining a student-school matching environment that utilizes the TTC mechanism. In the experimental design, we specifically focus on scenarios in which preferences conflict with priorities since these are the situations in which misreporting has been observed most often (Chen and Sönmez, 2006; Pais and Pintér, 2008). To remove any possible impact of strategic uncertainty, we cast the experiment in a decision setting. Using a traditional description of the TTC mechanism, subjects report their ordinal preferences truthfully in 22% of cases. Under the pick-the-best description, the truth-telling rate increases to 32%. The pick-the-best description increases the truth-telling rate by 9.6 percentage points. Hence the treatment is effective in improving the rate of truth-telling, but plenty of misreporting remains. We also measure subject numeracy using the Berlin Numeracy Test (BNT, Cokely, Galesic, Schulz, Ghazal and Garcia-Retamero, 2012). The effect of the pick-the-best description exhibits a strong positive interaction with numeracy. Under a traditional description, numeracy has no impact on the truth-telling rate, while under the pick-the-best description, numeracy has a strong positive impact on truth-telling. The truth-telling rate of low-numeracy subjects is little-changed by the treatment, while that of high-numeracy subjects increases substantially.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 introduces a theoretical framework that allows a formal statement of the proposed description and other results. Section 4 outlines the design of the experiment. Section 5 presents our empirical findings. Section 6 discusses the main findings. Finally, Section 7 concludes.

2 Literature Overview

Hassidim, Marciano, Romm and Shorrer (2017) identify four potential classes of explanations for why players do not report their types truthfully in strategy-proof mechanisms: cognitive limitations, mistrust in the designer, multiplicity of best responses and behavioral preferences. Our paper relates to the first one, namely cognitive limitations. Indeed, experimental subjects with lower cognitive ability have been found to be more likely to overbid in SPA (Lee, Nayga Jr, Deck and Drichoutis, 2020) and to misreport their ordinal preference under TTC (Guillen and Hakimov, 2017). Taken from a different angle, Guillen and Hakimov (2018) find that describing the TTC algorithm *reduces* dominant strategy play in comparison to just telling subjects that a mechanism they face is strategy-proof. One type of cognitive ability essential for recognizing strategy-proofness is the ability to think contingently. There is a growing literature pointing out challenges people might have with contingent thinking (Charness and Levin, 2009; Martínez-Marquina, Niederle and Vespa, 2019; Esponda and Vespa, 2023).

Another reason for misreporting is that players might not trust the mechanism designer to implement the mechanism as described or they might not trust a designer’s advice on how to play, if provided. For example, a bidder in SPA might be afraid that the auctioneer will try to exploit his bid by, after seeing all bids, entering an artificial shill bid just below the highest real bid.

Given that truthful reporting is a *weakly*, but often not a strictly dominant strategy, there might also be many misreporting strategies that are, given what a player believes about the other players' reports, payoff-equivalent to the truthful report. For example, if a student in a matching algorithm considers some schools to be impossible for him to get into, he might simply not rank them or rank them randomly.⁴ Or, if a bidder in SPA believes that the highest bid of the other bidders is at least d away from his value, overbidding or underbidding by less than d is payoff-equivalent to bidding truthfully.

Finally, players' preferences may differ from the pure material payoffs induced by the underlying game. For example, a bidder in SPA might overbid out of spite toward the winner (Morgan, Steiglitz and Reis, 2003; Andreoni, Che and Kim, 2007) or due to joy of winning (Cooper and Fang, 2008). A student might not rank his most preferred school on top of his list if he is not very likely to get admitted into it because of the disappointment that a rejection from that school would cause (Dreyfuss, Heffetz and Rabin, 2022b; Meisner and von Wangenheim, 2023; Dreyfuss, Glicksohn, Heffetz and Romm, 2022a; Chen, Katuščák, Kittsteiner and Kütter, 2023), or because it would hurt his ego (Köszegi, 2006). Also, a student might exhibit "rank-based utility" in that he likes having assigned a high rank to the school that he gets into (Meisner, 2023; Kloosterman and Troyan, 2023). Or, he might prefer a school that assigns him a high priority because he likes to be liked (Opitz and Schwaiger, 2023a,b). In the presence of such behavioral preferences, not ranking or bidding according to the material payoff consequences of allocations is not necessarily suboptimal. It is therefore unclear to what extent "misreporting" constitutes a problem that the designer might want to address.

Focusing on the cognitive limitations explanation, one approach to address the problem of mechanism complexity is to use an obviously-strategy-proof mechanism (OSP, Li, 2017) in place of "just" a strategy-proof (SP) mechanism.⁵ For example, it is arguably easier to see that truthful bidding is dominant in the English clock auction than it is in SPA. Or, likewise, in RSD, dominant strategy play is arguably easier to grasp in the dynamic implementation than in the simultaneous-choice implementation. Replacing a SP mechanism with its OSP version runs into two problems, however. First, many strategy-proof mechanisms, including DA or TTC, cannot in general be reformulated as an OSP mechanism (Li, 2017; Ashlagi and Gonczarowski, 2018; Troyan, 2019). Second, a shift from a SP to an OSP version of the mechanism changes

⁴See the analysis by Hassidim, Romm and Shorrer (2021) of data from Li (2017) from the lab and Shorrer and Sóvágó (2023) from the field for causal evidence that a decreased likelihood of being able to obtain a highly preferred option (because of an unlucky priority draw or because of increased competition, respectively) results in lower reported ranking of such option. Also, Artemov et al. (2023) document in a field setting that although applicants do submit what appear to be dominated rankings of study programs, such "mistakes" are in most cases payoff-irrelevant.

⁵A strategy is "obviously dominant" if the minimum possible payoff achieved by playing this strategy is at least as high as the highest possible payoff from any alternative strategy. In extensive form games, the comparison is performed at the node (or information set) in which an alternative strategy first deviates from the dominant strategy. The "obviousness" comes from the fact that no contingent thinking is necessary. All that is required is a comparison of payoff ranges of alternative strategies. A mechanism is obviously strategy proof if it is obviously dominant for each type of a player to report his type truthfully.

the underlying game form, which might change behavior of players with behavioral preferences for reasons other than complexity reduction, which might be undesirable. For example, a participant who is averse to disappointment might rank a less desirable but more obtainable object ahead of a more desirable but less obtainable one in the simultaneous implementation of RSD, while choosing in exactly the opposite way if presented with a hot choice between these two objects in the dynamic implementation of RSD. In contrast, our proposed description can be applied to any strategy-proof mechanism. Moreover, since our approach changes the description of the mechanism but not the underlying game form, it has a smaller potential to affect behavior of participants with behavioral preferences for reasons not connected to complexity reduction.

Our description shares some common features with prior theoretical work on characterizing outcomes of strategy-proof mechanisms as a result of decentralized decision making. The central idea is that, *ex post*, the allocation assigned to a participant can be characterized as her optimal choice from a menu or budget set. [Hammond \(1979\)](#) shows that, for exchange economies, the allocation a participant obtains is her most preferred allocation from a “menu” of possible allocations where that menu depends on the type reports of the other players and is independent from her own reported preferences. For DA and TTC, respectively, [Azevedo and Leshno \(2016\)](#) and [Leshno and Lo \(2021\)](#) characterize “budget sets” derived from the outcome of the matching procedure. A student’s match is the best school (according to her reported preferences) from her budget set. Different to the menus in [Hammond \(1979\)](#) and our feasible sets, these “budget sets” depend on a student’s reported type (and might not contain all feasible schools). Therefore, while the menus in [Hammond \(1979\)](#) are equivalent to our feasible sets, the budget sets in [Azevedo and Leshno \(2016\)](#) and [Leshno and Lo \(2021\)](#) are not. Such menus or budget sets can serve as an effective tool of *ex post* verification that one indeed obtained the best allocation for the submitted preference type from among the allocations that were available to her. In contrast, our approach stresses existence and usefulness of exogenous feasible sets *ex ante* for the purpose of describing the mechanism to the player in a way that enhances the perception of strategy-proofness. [Meisner \(2023\)](#) uses the concept of feasible sets (his “attainability states” are equivalent to our feasible sets) to identify properties of distributions over feasible sets (generated by some strategy-proof matching environment) for which misreporting is (sub)optimal if a student’s utility is report-dependent.

Whereas we describe the construction of feasible sets in the most general way applicable to any strategy-proof mechanism, [Gonczarowski, Heffetz and Thomas \(2023\)](#) provide computationally less-demanding and arguably simpler descriptions of the construction of feasible sets (they call “menus”) for specific matching mechanisms such as DA. They also test the effectiveness of the two-step description for increasing truth-telling in median voting and SPA (see [Section 6](#) for further details).

Our description highlights information that suggests that reporting truthfully is an optimal strategy. In that sense, our study is related to the literature on the effects of providing advice in strategy-proof matching mechanisms. Whereas [Guillen and Hing \(2014\)](#) find in a lab experi-

ment that advice has a negative effect on the rate of truth-telling, [Braun, Dwenger, Kübler and Westkamp \(2014\)](#) and [Ding and Schotter \(2019\)](#) in lab experiments and [Guillen and Hakimov \(2018\)](#) in a field experiment find that advice increases the rate of truth-telling. Our setup resembles that in [Guillen and Hakimov \(2018\)](#) in two important aspects. The information that is central to inferring the strategy-proofness property (i.e., the advice in [Guillen and Hakimov \(2018\)](#) and the two steps in our description) is coming from a supposedly trustworthy source (the lecturer or the experimenter, respectively) and this information is separated from the technical details of the mechanism description (where [Guillen and Hakimov \(2018\)](#) do not provide any description of the mechanism in their most effective treatment).

3 Theory

In this section, we formalize the underlying economic environment and demonstrate that any strategy-proof, and only a strategy-proof, direct-revelation mechanism can be framed as a situation where a player's report is used to pick a best, according to the reported type, allocation from a set of feasible allocations that is independent of that report. We then describe how this theoretical result can be converted into a two-step description of any strategy-proof mechanism for the purpose of enhancing the perception of strategy-proofness and use the student-school matching environment as an example to demonstrate general concepts and results presented in this section.⁶

3.1 Setup

We consider a private values environment with $i = 1, \dots, n$ players who need to make a collective choice from a set of possible allocations $\mathcal{X}$. Prior to making a decision, each player observes her type θ_i that uniquely determines her (weak) *ordinal* preference relation over $\mathcal{X}$. This preference relation is represented by a utility function $u_{\theta_i}(\cdot)$. The set of possible types of player i is denoted by Θ_i . We do not make any assumptions regarding the distribution of types but assume that the sets of possible types $\Theta_i, i = 1, \dots, n$ are common knowledge.

For breaking ties and potentially other type-independent (random) inputs into determining the allocation (e.g., school priorities in student-school matching), we consider a randomization device with a commonly known distribution over its possible realizations λ . We label the set of possible realizations by Λ . λ is known to the mechanism implementer, but not necessarily to each player. Let $z_i : \Lambda \rightarrow \Lambda_i$ be a mapping that captures what player i knows about λ , with $\lambda_i \equiv z_i(\lambda) \in \Lambda_i$ being her information set and Λ_i being the set of her possible information sets.

Given a social choice function $g : \Theta_1 \times \dots \times \Theta_n \times \Lambda \rightarrow \mathcal{X}$, we consider a direct revelation mechanism $\Gamma = (\Theta_1, \dots, \Theta_n, \Lambda, g)$. The sequencing of events is as follows. First, Nature draws $\lambda \in \Lambda$. The realized λ is revealed to the mechanism implementer and λ_i is revealed to player

⁶In Appendix A, we also demonstrate the concepts and results using the Vickrey auction example.

i . Next, each player i independently reports a type $t_i \in \Theta_i$.⁷ Finally, a deterministic allocation $g(t_1, \dots, t_n, \lambda)$ is chosen. In what follows, we denote by θ_{-i} and θ the tuple of true types of all the other players but i and of all the players, respectively. Similarly, we denote by t_{-i} and t the tuple of reported types of all the other players but i and of all the players. Likewise, we will use Θ_{-i} and Θ to denote sets of types of all players other than i and of all the players, respectively. A tuple $(t_{-i}, \lambda) \in \Theta_{-i} \times \Lambda$ is referred to as a *contingency* (for player i).

We restrict attention to strategy-proof mechanisms.

Definition 1. A mechanism Γ is **strategy-proof** if for all $i = 1, \dots, n$, all $\theta_i \in \Theta_i$ and all contingencies $(t_{-i}, \lambda) \in \Theta_{-i} \times \Lambda$ it is the case that

$$u_{\theta_i}[g(\theta_i, t_{-i}, \lambda)] \geq u_{\theta_i}[g(t_i, t_{-i}, \lambda)]$$

for all $t_i \in \Theta_i$.

To simplify the environment from a specific player's perspective, we cluster all allocations among which, independently of her type, the player is indifferent. We will subsume all such allocations under a new label, which we refer to as a *private allocation*.

Definition 2. A set $\mathcal{X}_i$ is called a set of **private allocations** of player i if there exists a mapping $h_i : \mathcal{X} \rightarrow \mathcal{X}_i$ such that for any $x_i \in \mathcal{X}_i$ it is the case that:

1. any type $\theta_i \in \Theta_i$ of player i is indifferent among all allocations in $h_i^{-1}(x_i)$;
2. there is no set $B \subset \mathcal{X}$ such that $h_i^{-1}(x_i) \subset B$ and every type $\theta_i \in \Theta_i$ of player i is indifferent among all allocations in B (i.e., allocations are bundled together as much as possible for the purpose of constructing private allocations).

The concept of the set of private allocations does two things. First, it implicitly partitions $\mathcal{X}$ into largest possible type-independent indifference sets. Second, it assigns to every such indifference set a (convenient) label. Note that whereas the partition is uniquely determined by the fundamentals of the problem (and is well defined), the labels can be chosen at will.

Given $\mathcal{X}_i$, the *private allocation function* of player i is given by $g_i : \Theta \times \Lambda \rightarrow \mathcal{X}_i$ with $g_i(t, \lambda) \equiv h_i[g(t, \lambda)]$. Let v_{θ_i} denote the utility function induced by u_{θ_i} on $\mathcal{X}_i$, i.e., for any $x_i, y_i \in \mathcal{X}_i$ and any $x \in h^{-1}(x_i)$ and $y \in h^{-1}(y_i)$, we have that $v_{\theta_i}(x_i) \geq v_{\theta_i}(y_i) \Leftrightarrow u_{\theta_i}(x) \geq u_{\theta_i}(y)$.

To illustrate these theoretical concepts, consider the following student-school matching example.

Matching example: n students (the players) are to be assigned to m schools from the set $S \equiv \{s_1, \dots, s_m\}$, or to be left unassigned, denoted by s_0 . School s_p has n_p available seats. The set of allocations is $\mathcal{X} = \{(y_1, \dots, y_n) : y_i \in \{s_0, s_1, \dots, s_m\}, i = 1, \dots, n, |\{j : y_j = s_p\}| \leq$

⁷Note that since the mechanism implementer knows λ , which is a sufficient statistic for λ_i , λ_i is not a part of player i 's privately known type that needs to be elicited by the mechanism.

$n_p, p = 1, \dots, m\}$. In this notation, y_i is either the school student i is assigned to, or s_0 , if the student is unassigned. A player i 's type θ_i is given by her strict preference ordering over $\{s_0, s_1, \dots, s_m\}$. This assumes away any non-pecuniary allocation externalities such as peer effects. Then the set $\mathcal{X}_i$ of private allocations of player i is defined by $\mathcal{X}_i \equiv \{s_0, s_1, \dots, s_m\}$ with

$$h_i(y_1, \dots, y_n) \equiv y_i.$$

We can choose

$$v_{\theta_i}(y_i) = u_{\theta_i}(y_1, y_2, \dots, y_n).$$

Assuming that schools' priorities over students are non-strategic, some of the best-known allocation functions (see Section 1) are generated by DA, TTC or (R)SD. λ captures the school priorities. In the most general case, Λ is the cross-product of m sets of all $(n+1)!$ permutations of the set $\{0, 1, \dots, n\}$. For any $\lambda = (\lambda^1, \dots, \lambda^m)$, $\lambda^p(i)$, $i \in \{0, 1, \dots, n\}$ is the position of student i or the dummy index 0 in the permutation of school p . Any student i with $\lambda^p(i) < \lambda^p(0)$ is acceptable for school p and his or her priority in that school is given by $\lambda^p(i)$, with an earlier position in the permutation indicating a higher priority. On the other hand, any student i with $\lambda^p(i) > \lambda^p(0)$ is deemed unacceptable for school p . ■

To state our main observation, we need to introduce the concept of feasible sets.

Definition 3. Given a strategy-proof mechanism Γ with a private allocation function g_i and a contingency $(t_{-i}, \lambda) \in \Theta_{-i} \times \Lambda$, the **feasible set of player i** is given by

$$F_i(t_{-i}, \lambda) \equiv \{x_i : \exists t_i \text{ with } x_i = g_i(t_i, t_{-i}, \lambda)\}.$$

Let

$$\mathcal{F}_i \equiv \{F_i(t_{-i}, \lambda) : (t_{-i}, \lambda) \in \Theta_{-i} \times \Lambda\}$$

be the collection of all possible feasible sets of player i . Also, for any $f_i \in \mathcal{F}_i$, let

$$\tau_{\theta_i}(f_i) \equiv \{x_i \in f_i : v_{\theta_i}(x_i) \geq v_{\theta_i}(\tilde{x}_i) \text{ for all } \tilde{x}_i \in f_i\}$$

be the set of private allocations in f_i that type θ_i of player i prefers the most. Note that since player i can report truthfully under any contingency and can never do better, $g_i(\theta_i, t_{-i}, \lambda) \in \tau_{\theta_i}(f_i)$ for any contingency $(t_{-i}, \lambda) \in F_i^{-1}(f_i)$. As a result, $\tau_{\theta_i}(\cdot)$ is non-empty. Finally, let $V_{\theta_i} : \mathcal{F}_i \rightarrow \mathbb{R}$ defined by

$$V_{\theta_i}(f_i) = v_{\theta_i}(x_i) \text{ for any } x_i \in \tau_{\theta_i}(f_i).$$

for any $f_i \in \mathcal{F}_i$ be the associated “value function.”

3.2 Theoretical Results

Our proposed description of *any* strategy-proof mechanism is based on the following Proposition.

Proposition 1. *A mechanism Γ is strategy-proof if and only if for any $(t_i, t_{-i}, \lambda) \in \Theta \times \Lambda$ and any i , it allocates to player i a private allocation that is among the most preferred private allocations of type t_i in $F_i(t_{-i}, \lambda)$, that is*

$$g_i(t_i, t_{-i}, \lambda) \in \tau_{t_i}[F_i(t_{-i}, \lambda)]. \quad (1)$$

Proof. Suppose Γ is strategy-proof. By construction of feasible sets, $g_i(t_i, t_{-i}, \lambda) \in F_i(t_{-i}, \lambda)$. Also, by Definition 1, for any type $t_i \in \Theta_i$ of player i and given any $(t_{-i}, \lambda) \in \Theta_{-i} \times \Lambda$,

$$v_{t_i}[g_i(t_i, t_{-i}, \lambda)] \geq v_{t_i}[g_i(t'_i, t_{-i}, \lambda)]$$

for all $t'_i \in \Theta_i$. The two observations together imply (1). In the opposite direction, (1) implies that there is nothing to gain for player i from misreporting her type, meaning that Γ must be strategy-proof. $\square$

Proposition 1 is central to the idea of this paper. It shows that, from the point of view of each player, any strategy-proof mechanism can be described as a problem to pick (one of) the best feasible allocation(s). When asked to submit her type, a player faces an unknown set of private allocations, namely her feasible set given the reports of the other players (and λ). After all the players have submitted their types and λ has been determined, her feasible set is computed and the preference order implied by her reported type is then used to choose (one of) the most preferred private allocation(s) from her feasible set.

A version of Proposition 1 for exchange economies was formulated by Hammond (1979, Theorem 1). His interpretation, however, is different from ours in that he focuses on the *ex-post* nature of the decentralization result. More specifically, he concludes that, given the type reports of the other players, one could announce the feasible set (“menu” in his labelling) to a player and let the player choose an optimum for himself. For practical applications, though, such a description is only possible for one of the players as the other players’ reports are needed beforehand to construct his feasible set. In contrast, we concentrate on the implications of Proposition 1 for a simultaneous *ex-ante* reformulation of a strategy-proof mechanism to any particular player.

Note that if any type of player i can only have strict preferences over private allocations, then, given t_i , $F_i(t_{-i}, \lambda)$ is sufficient to determine $g_i(t_i, t_{-i}, \lambda)$. This is because $\tau_{t_i}[F_i(t_{-i}, \lambda)]$ is a singleton and hence

$$g_i(t_i, t_{-i}, \lambda) = \tau_{t_i}[F_i(t_{-i}, \lambda)].$$

To illustrate, consider the matching example.

Matching example: Assuming that each type of each player can have only strict preferences over $\{s_0, s_1, \dots, s_m\}$, then $\tau_{\theta_i}(f_i)$ is a singleton for any possible feasible set f_i and any type θ_i . As a result, f_i is sufficient to determine the private allocation of player i for any report t_i he might submit. No information from λ is needed for this purpose. ■

If some types of player i have indifferences over private allocations, then $\tau_{t_i}[F_i(t_{-i}, \lambda)]$ is not necessarily a singleton and hence, given t_i , $F_i(t_{-i}, \lambda)$ is sufficient only for determining a (restricted) set from which $g_i(t_i, t_{-i}, \lambda)$ is chosen, but not necessarily for determining $g_i(t_i, t_{-i}, \lambda)$ itself. However, since $v_{\theta_i}(x_i)$ has the same value for all $x_i \in \tau_{\theta_i}[F_i(t_{-i}, \lambda)]$, $F_i(t_{-i}, \lambda)$ is sufficient for determining the payoff (welfare) of player i if this player reports truthfully. These observations are illustrated in our Vickrey auction example in Appendix A.

To determine a player's feasible set, the designer in principle needs to run the mechanism for each possible report of the player's type, given the actually reported types of the other players and the realization of λ . For example, in the matching setting, constructing a player's feasible set in principle requires submitting all possible rankings over $\{s_0, s_1, \dots, s_m\}$ against the given contingency. However, the following corollary to Proposition 1 shows that in some settings, for example in the matching setting, the problem is actually quite manageable, as $F_i(t_{-i}, \lambda)$ can be constructed by top-ranking, one-by-one, all $x_i \in \mathcal{X}_i$, ranking the other private allocations below arbitrarily.

Corollary 1. *If, for each player i , $\mathcal{X}_i$ is finite and Θ_i consists of all strict preference orders on $\mathcal{X}_i$, then, for any contingency $(t_{-i}, \lambda) \in \Theta_{-i} \times \Lambda$, $x_i \in F_i(t_{-i}, \lambda)$ if and only if $g_i(t_i, t_{-i}, \lambda) = x_i$ whenever x_i is the highest-ranked private allocation in t_i .*

Proposition 1 also offers insights into the concept of partition-obvious dominance of Zhang and Levin (2021). This connection is formally explored in Appendix B.

3.3 “Pick-the-best” mechanism description

The results and the discussion from the previous subsection suggest that any strategy-proof mechanism can be described to any player i as the following two-step procedure (here presented in technical jargon, to be presented non-technically in our application examples below):

1. “Based on the reported types of the other players (and the priority orderings), the mechanism determines which private allocations you can feasibly get. *Your reported type plays no role in this step.*”
2. “The mechanism then selects for you a feasible private allocation that you prefer the most according to your reported type.”

The players are also referred to a description of how the set of feasible allocations is constructed for the given contingency in step 1, but this information is de-emphasized. This description might also explain how, in case of a tie, one of the multiple most preferred allocations is chosen in step 2.

To illustrate how this description could look like in a particular application, consider our matching example.

Matching example:

1. Based on the reported rankings of the other players, the mechanism determines which schools are feasible to you. You can get into any of these schools with certainty and you will not be able to get into any other school. *This step does not use your reported ranking in any way.*
2. The mechanism then selects for you the feasible school from step 1 that is highest up in your reported ranking. If no school is feasible for you or if you rank being unassigned higher than any feasible school, you will not be assigned to any of the schools. ■

An important design question is how to describe the construction of feasible sets in Step 1. First, regarding salience of this information, our preferred choice is to make it *optionally* available via an appendix. This makes the information available to those who want it, while emphasizing only the elements of the mechanism that are sufficient to infer strategy-proofness in the main text. Second, regarding the construction itself, it might be described in several ways, depending on the application. As discussed in the previous section, a universal way to describe the construction of the feasible set for a player is to consider submitting all possible reports for this player and collect the resulting private allocations. In some environments, such as matching, the description can be simplified by considering only all possible one-school truncated ranking reports (see Corollary 1). This is what we use in the experiment presented in this paper. In specific matching mechanisms, it might be possible to simplify the description of the construction of feasible sets even further. For example, [Gonczarowski et al. \(2023\)](#) show that, in DA, the feasible set can be constructed by running the school-proposing version of DA, omitting the student in question, and identifying her feasible set by all the schools in which the priority of the lowest-priority admitted student is below the priority of the student in question. The exploration of the effect of such simplifications, if available, is left to future research.

4 Experimental Design

4.1 Setting

The experimental design aims to test whether the pick-the-best description increases the rate of truthful type reporting. We focus on the matching application since ties do not need to be dealt with, simplifying the experiment. We use TTC as the underlying mechanism. We prefer this mechanism to DA for two reasons.⁸ First, we believe that a traditional description of TTC is more complex (i.e., more difficult to understand and execute computationally) than one of DA. TTC involves formation of cycles that do not have a parallel in real life that subjects would be

⁸For additional reasons why TTC is a good choice, see [Guillen and Hakimov \(2018\)](#).

familiar with. DA, on the other hand, might be more familiar to some subjects as it resembles real life matching situations in which provisional assignments (until a better match shows up) can occur. Also, in TTC, final allocations of other players can occur before the own match is finalized. This might create additional confusion about whether a player can miss out on a less preferred school while “chasing” the most preferred one. Thus, we see more potential for correction of misreporting in TTC than in DA. Second, in TTC, unlike in DA, feasibility of a particular school depends on the applicants’ priorities in all schools, not just the school in question. This implies that, other things equal, priorities in the individual schools are less predictive of probabilities of feasibility of those schools in TTC in comparison to DA. As a result, payoff consequences of deviating from truth-telling in the direction of one’s priorities are typically larger, implying that whether one does vs. does not report truthfully is more payoff-relevant.

The experiment is based on a simple setting with four “applicants” and four “institutes,” each having one available position. Because we want to abstract from strategic uncertainty and, potentially, social preference concerns, we recast the game as a decision situation. In particular, in the group of four applicants, one subject interacts with three computerized applicants.⁹ The computerized applicants submit random rankings of the four institutes. Each of them randomizes independently and submits each of the 24 possible rankings with an equal probability. The human applicant’s ranking is her decision. Each institute has a priority ordering over the four applicants. An applicant’s priority in an institute is an integer between 1 and 4, with 1 representing the highest priority and 4 representing the lowest priority.

Given the applicants’ submitted rankings and the institutes’ priority orderings, the TTC mechanism proceeds as follows. In iteration 1, each applicant “points” to her highest-ranked institute, while each institute “points” to its highest-priority applicant. In this directed graph with finitely many nodes, there must be at least one cycle. Also, each cycle participant (an applicant or an institute) can be in at most one cycle. Within any such cycle, each applicant is allocated the position in the institute she is pointing to. Such applicant-institute pairs are then removed. In iteration 2, the procedure is repeated with the remaining applicants and institutes. The difference is that each applicant now points to her highest-ranked *remaining* institute and each institute now points to its highest-priority *remaining* applicant. Whichever applicants and institutes remain unassigned at the end of iteration 2 then proceed to iteration 3 that works analogously to iteration 2. And so on until all applicants are assigned to a position. Given the number of applicants and institutes, there are at most four iterations. At the end of the procedure, each applicant is assigned to one institute. There is no option to remain unassigned in our setting.¹⁰

⁹In this respect and also in terms of the number of applicants and positions, our design is similar to that of [Guillen and Hakimov \(2017\)](#) or [Klijn, Pais and Vorsatz \(2019\)](#).

¹⁰Note that this mechanism description differs from that in other experiments in the literature which makes a direct comparison of truth-telling rates to their setups difficult. Most notably, [Chen and Sönmez \(2006\)](#), due to each student having a district school which guarantees him a seat (translated into the language of our setup, in

For expositional purposes, we label the four institutes A , B , C and D , from the highest- to the lowest-payoff for the human applicant. In the experiment itself, we label them by neutral names: Circle, Diamond, Square and Triangle. The mapping between these labels and our exposition labels A , B , C and D is randomized.

We induce subject preferences over the four institutes as follows. The point payoff for acquiring the position in institute A is chosen randomly, with equal probabilities, from the set $\{91, 92, \dots, 99\}$; that for institute B from the set $\{61, 62, \dots, 69\}$; that for institute C from the set $\{31, 32, \dots, 39\}$; that for institute D from the set $\{1, 2, \dots, 9\}$. We use the exchange rate 20 points = €1.

Before submitting their ranking, subjects are, apart from their values, also informed about their priorities in each of the four institutes. With a , b , c and d being the respective priorities in institutes A , B , C and D , we refer to the sequence $abcd$ as the applicant's *priority profile*.

We would ideally like to measure whether or not subjects correctly perceive the strategy-proofness of the mechanism. Although we do ask subjects *ex post* to comment in a free form on how they decided on their rankings, which is to at least some degree indicative of their mechanism perception, we primarily rely on the submitted rankings, which are noisy signals of whether the strategy-proofness is perceived correctly or not. While submitting a non-truthful ranking can arguably be considered as a strong signal of not perceiving the strategy-proofness correctly, submitting a truthful ranking is not necessarily as strongly indicative of perceiving the strategy-proofness correctly. This is because even someone who does not perceive the strategy-proofness correctly might report the true ranking just because the priority profile does not contradict perceived optimality of such report in any way. In particular, if $a \leq b \leq c \leq d$, then reporting the true ranking $ABCD$ is most likely the focal choice irrespective of whether the mechanism is perceived to be strategy-proof or not. For example, given such priority profile, the truthful report is optimal also in the immediate acceptance (or Boston) mechanism. Hence in order to increase the chance that not perceiving the strategy-proofness translates into misreporting the true preference, we create a conflict between preferences and priorities. Previous experimental literature ([Chen and Sönmez, 2006](#); [Pais and Pintér, 2008](#)) suggests that such conflict is conducive to misreporting. In particular, we exploit what this literature has labeled the “district school bias”. This refers to a situation in which an applicant has a relatively high priority in a school that is quite high up her true ranking (the “district school”), while her priority in her most preferred school is lower (or low). In such situation, an applicant might feel that “a bird in the hand is worth two in the bush” and top-rank her district school at the expense of her truly most preferred school. Indeed, such preference manipulation might be optimal in the immediate acceptance mechanism since the district school might fill up while one is chasing her top-preferred school. However, such preference manipulation is not optimal *ex ante* in a

which he has the priority 1), can explain the creation of cycles in the TTC mechanism by using the notion of an “initial tentative assignment” to his district school. In our setting with students who have no or multiple district schools, we cannot do that.

Round 3

Enter your ranking in the input fields!

Institute	Value	Your priority in the institute	Your rank for the institute
Square Institute	3	2	Enter 1, 2, 3 or 4!
Circle Institute	66	3	Enter 1, 2, 3 or 4!
Diamond Institute	32	1	Enter 1, 2, 3 or 4!
Triangle Institute	92	3	Enter 1, 2, 3 or 4!

[Continue](#)

Figure 1: Screenshot of the decision screen (translated into English)

strategy-proof mechanism as long as there is a positive probability that the two schools are both feasible (i.e., are both in the feasible set). Building up on this idea, we design the decision scenarios (see the next subsection) such that institute B gives a higher priority to the human applicant than institute A does. Secondly, we sometimes also create a similar conflict between preferences and priorities for other pairs of institutes.

4.2 Scenarios

In order to collect more information on individual responses to various priority profiles, each subject submits his ranking for 20 different scenarios. These are presented sequentially and labeled to subjects as “rounds.” Figure 1 presents a screenshot of the decision screen for one such round. For each of the four institutes, a human applicant observes her induced value in points and her priority in this institute. She is then asked to submit her ranking by filling in the numbers 1 (the highest rank) through 4 (the lowest rank) into the four boxes corresponding to the individual institutes. The choice must then be confirmed. The order in which the four institutes are presented on the decision screen is randomized.

Of the 20 scenarios, 18 create a conflict between preferences and priorities as described in the previous subsection. We will refer to these scenarios as *conflict* scenarios. Table 1 presents the design of the priority profiles in the 18 conflict scenarios. The subject has a higher priority in Institute B than she has in institute A , and the design spans all 6 possibilities with $a > b$. Regarding institutes C and D , we explore all three options: $c > d$, $c < d$ and $c = d$. These possibilities are then combined into $6 \times 3 = 18$ priority profile scenarios.

The two remaining scenarios, that we refer to as *placebo* scenarios, have flat priority profiles 2222 and 3333. We include these placebo scenarios in order to check the extent of truth-telling

Table 1: Design of priority profiles in the 18 conflict scenarios

		Priorities in A and B : $ab..$					
		41..	31..	42..	32..	21..	43..
Priorities	..11	4111	3111	4211	3211	2111	4311
in C and D :	..31	4131	3131	4231	3231	2131	4331
$..cd$	..34	4134	3134	4234	3234	2134	4334

Note: The four-digit sequences refer to a priority profile $abcd$ in which a , b , c and d are the priorities in institutes A , B , C and D , respectively.

when priorities do not conflict with preferences. The extent to which subjects do not report the truth in these two scenarios can be understood as the extent of subject confusion and noise in the data.

Conditionally on the designed priority profile of the human applicant, the rest of the priority orderings of each institute are generated randomly, with all six possible permutations being equally likely, independently across the institutes. Subjects are made aware of this in the instructions.

The 20 scenarios are presented in a randomized order as rounds and without any round-to-round feedback. The feedback is only presented at the end of the experiment. All the required randomizations are performed independently across rounds. Subjects are paid based on the sum of their point earnings in 3 randomly chosen rounds.

4.3 Treatments

We implement two description treatments. In the *Traditional* treatment, the mechanism is described in a “classical” way. The instructions consist of one printed document. In the *Pick-the-best* treatment, the mechanism is described using our proposed pick-the-best description. The instructions consist of two printed documents: the main instructions and an appendix describing how the feasible set is constructed from the submitted rankings of the three computerized applicants and the realized priority orderings of the four institutes. The two documents are clearly labelled. The main instructions refer to the appendix, but do not ask subjects to study it. In both treatments, the first part of the (main) instructions (4 pages) is identical. It describes the environment, the available information and provides a screenshot to familiarize subjects with the decision interface (this screenshot is reproduced in Figure 1). In *Traditional*, the next page then describes the TTC procedure. In *Pick-the-best*, it describes the two-step procedure.

Both the (main) instructions and the appendix contain two examples, identical across the two treatments. The examples start from assumed rankings submitted by the subject and by the three computerized applicants and assumed priority orderings. In *Traditional*, the examples walk subjects through the cycle formation steps as described above until a final allocation is

reached. The descriptions are complemented by graphical depictions of the cycles. In *Pick-the-best*, the very same examples are split into two parts. In the appendix, they walk subjects through the construction of the feasible set (including the graphical depictions), labelled to subjects as “the set of institutes available to you.” In the main instructions, this set is then used to demonstrate how the assumed ranking of the human applicant is used to choose the highest-ranked institute from the feasible set. The two examples are constructed such that the number of feasible institutes is 2 in one of them and 3 in the other one.

The instructions are, after translation from German to English, reproduced in Appendix D.

4.4 Additional data collection

After completing the 20 rounds, we ask subjects to comment in a free form on how they made their decisions. In *Pick-the-best*, we also ask whether they have read the appendix and, if yes, whether they found it useful. We also ask about fluency in German.

Next, we administer the *Berlin Numeracy Test* (Cokely, Galesic, Schulz, Ghazal and Garcia-Retamero, 2012). This test is designed to measure cognitive abilities related to numeracy and risk literacy. Compared to other commonly used tests for cognitive ability (Cognitive Reflection Test, Raven’s matrices), this test has been shown to improve psychometric discriminability among highly-educated individuals such as college students, from among whom we draw our subject pool. The test consists of four multiple-choice questions, each with four possible answers. The questions inquire about probabilistic reasoning that by itself requires an ability to think contingently. We therefore speculate that this test is also a good proxy for the ability to think contingently. In order to incentivize the answers, subjects are paid 10 points (€0.50) for each correct answer. Most subjects complete the test in a few minutes.¹¹ The questions and the available answers are reproduced in Appendix E. The reason for collecting the numeracy responses is that we want to examine how the impact of the pick-the-best description on truth-telling interacts with cognitive ability.

Once done with the numeracy test, we run a short demographic questionnaire. We collect data on gender, age, subject of study and year of study.

4.5 Logistics

We compare the two treatments using a between-subject design. Each session is designed for 30 participants, half in each treatment. These constitute 15 matched pairs. We exactly replicate the realizations of the value draws, priority profiles, institute labels, institute presentation order on the decision screen and round ordering within each pair. This is intended to reduce noise potentially affecting comparison of the two treatments.

Our original aim was to have approximately 100 subjects per treatment. With this number of independent observations in each treatment, we are able to identify a treatment effect of a

¹¹The slowest ones might take up to 10 minutes to complete the task.

size of 40% of a standard deviation in the outcome variable with a power of 0.8. The outcome variable is the subject-specific fraction of truthful reports in the 18 conflict scenarios.

We conducted seven sessions in June 2018 in AIXperiment laboratory at RWTH Aachen University. All subjects were recruited using the *Online Recruitment System for Economic Experiments* (Greiner, 2015) from the subject database of the lab that consists mostly of students from RWTH Aachen University. After subjects check-in for the experiment, they draw a card with the workstation number and proceed to be seated there. This assures randomization in treatment assignment within each session. Due to subject no-shows despite over-recruitment, we had two missing subjects in sessions 1 and 4 and four missing subjects in session 6. In the latter session, we managed to pair the 26 subjects into 13 pairs. To obtain the missing data points for sessions 1 and 4, we added the four missing subject set-ups to session 7, in which 26 subjects participated altogether. In total, we have data for 99 subjects in each treatment, paired into 99 data pairs for noise reduction. Of the 198 subjects, 54% are men and 46% are women, and 95% of subjects are between the ages of 18 and 30.

After subjects are seated at their workstations, they are welcomed by an experimenter and prompted to read the printed instructions at their own pace. When it appears that most subjects have finished reading the instructions, the experimenter prompts them to privately ask questions if they have any. In case there are questions, they are privately addressed by the experimenter. There is no quiz or practice rounds. The reason is that we want to examine how the description itself, without any impact from feedback-driven potential learning, affects the reported rankings.

Subjects then proceed through the 20 rounds at their own pace. The experiment is computerized via *oTree* (Chen, Schonger and Wickens, 2016). In each round, subjects submit a ranking and confirm before being able to move to the next round. Once they have completed the 20 rounds, we collect additional data as described in Subsection 4.4. Once all that is completed, subjects are given feedback on their decisions in the 20 rounds. For each round, they are reminded of their values and the submitted ranking. In *Pick-the-best*, they are also informed of which institutes were feasible. They are then informed which institute they were assigned to in that round and the resulting point earnings. Subjects are also informed which three rounds are chosen to be payoff-relevant. They are also informed about their earnings from the numeracy elicitation tool, their final point earnings and their resulting cash earnings. As a reminder, we use the exchange rate 20 points = €1.

5 Results

5.1 Treatment effect

We refer to a submitted ranking in a given round as a *report*. We define truth-telling by providing the report $ABCD$.¹² In the conflict scenarios, the average truth-telling rate is 22.2% in *Traditional* and 31.8% in *Pick-the-best*. The treatment effect is 9.7 percentage points and it is statistically significant ($p = 0.025$).¹³ These truth-telling rates are rather small in comparison to the previous experimental literature.¹⁴ This could be driven by a smaller share of conflict scenarios among all scenarios in the previous literature. In support of this hypothesis, we find that the truth-telling rate in the placebo scenarios is 84.8% in *Traditional* and 87.9% in *Pick-the-best* (statistically not different). The stark difference in truth-telling rates between the conflict and the placebo scenarios suggests that, within our setting, one could obtain almost any average total truth-telling rate between 20% and almost 90% by suitably setting proportions of the two types of scenarios.

We consider the treatment to be effective on a desired margin if it leads to an increase in truth-telling by increasing the share of subjects who correctly perceive strategy-proofness of the mechanism. In order to get a more precise gauge of the extent to which the treatment effect operates via the intended channel, we conduct two types of analyses. First, a correct perception of mechanism strategy-proofness should manifest itself via a high fraction of truthful reports in the 18 conflict scenarios. For example, reporting truthfully in 9 out of these 18 scenarios is very unlikely to be consistent with the correct perception. On the other hand, reporting truthfully in all 18 scenarios is very unlikely to be driven by anything else except for the correct perception.¹⁵ Allowing for some reporting errors due to a lack of attention or “fat fingers,” even 17, 16 or 15 truthful reports might be considered to suggest the correct perception. Figure 2 displays incidence of such behavior across the two treatments. We observe that the share of subjects with 18 or near 18 truthful reports is much higher in *Pick-the-best* compared to *Traditional*. Regarding subjects who report truthfully in all 18 (17, 16 or 15) conflict scenarios, only 2% (4%, 4%, 4%) do so in *Traditional*, whereas 12.1% (14.1%, 17.2% and 18.2%) do so in *Pick-the-best*. The difference(s) are highly statistically significant (p -values ranging from 0.001

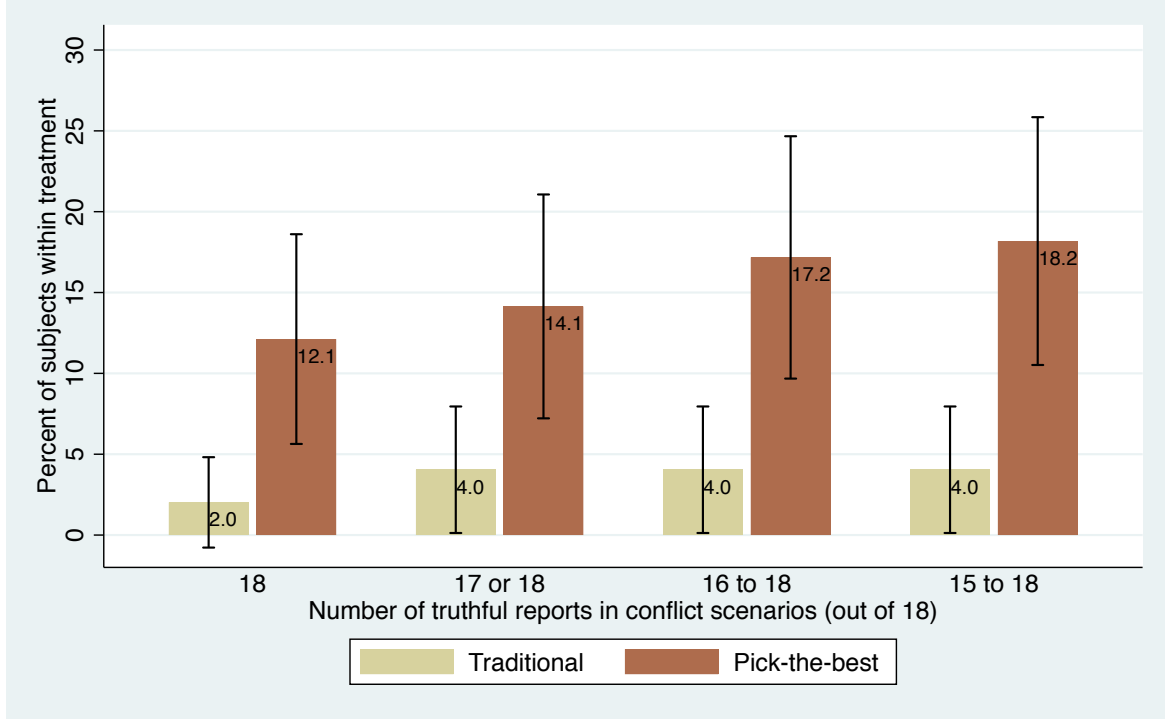
¹²For someone who understands the mechanism, submitting the report $ABDC$ while having priority 1 in institute B is payoff equivalent to submitting $ABCD$ irrespective of the underlying contingency. One could therefore argue that this case corresponds to truth-telling as well. For example, [Chen and Sönmez \(2006\)](#) consider such report to be truthful. However, it is disputable whether such report can be considered truthful in *Pick-the-best* if an applicant does not know details of step 1, i.e., how the feasible set is constructed. Since understanding details of how the feasible set is constructed is not necessary for correctly perceiving strategy-proofness, we do not consider this special case to be truth-telling. Empirically, this distinction is almost immaterial. Out of the 3960 reports, only 9 (or 0.23%) fit the particular data pattern.

¹³This and analogous tests presented below are two-sided tests of mean equality with standard errors adjusted for clustering at subject level.

¹⁴See [Hakimov and Kübler \(2021\)](#) and [Rees-Jones and Shorrer \(2023\)](#) for recent surveys.

¹⁵The only other potential explanation we can think of is subject confusion combined with reporting using preferences as the focal point. But if a subject uses preferences as opposed to priorities to focus on, that on its own might suggest some understanding of the strategy-proofness.

Figure 2: Incidence of a high count of truthful responses in the conflict scenarios (with 95% confidence intervals) by treatment



to 0.013). These observations suggest that the treatment effect on the average truth-telling rate is indeed driven by an increased perception of the strategy-proofness property.

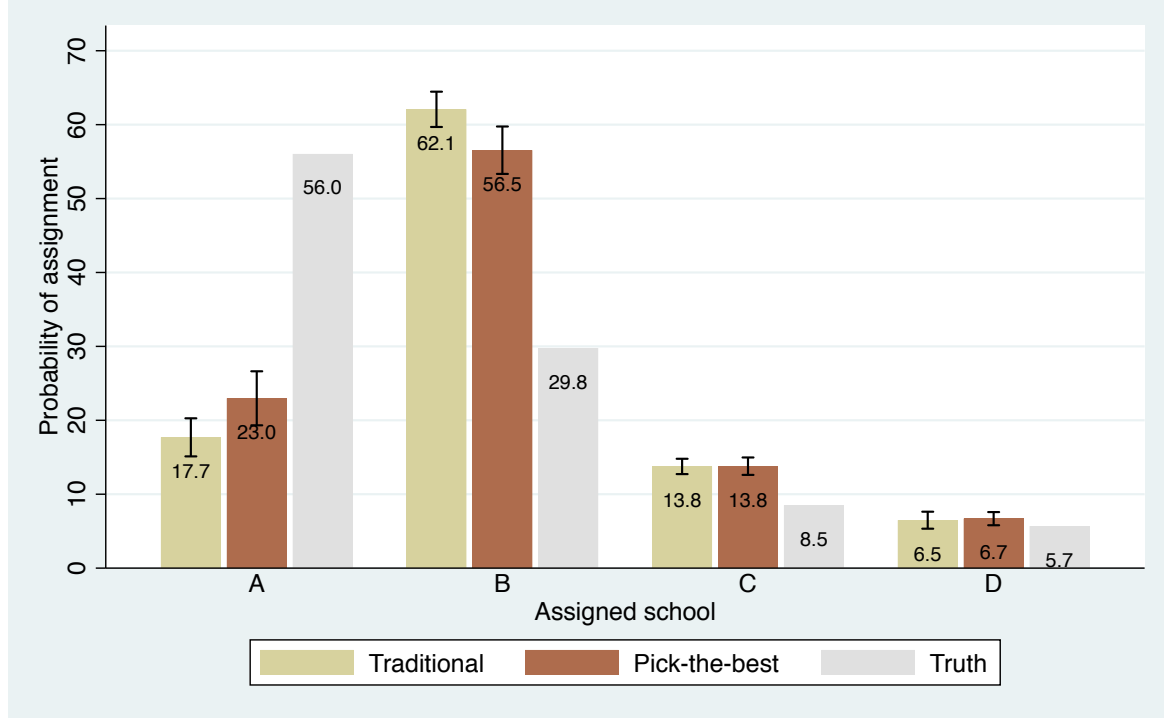
Second, instead of looking at the reports, we turn to subject comments. These comments provide an alternative source of data on intended reports, at least as of the end of the experiment. Recall that after having submitted their reports, subjects are asked to comment in a free form on how they approached what reports to submit. We categorize each comment into one of three categories: *strategy-proofness* (clearly stating one has followed induced preferences only), *trade-off* (clearly expressing following priorities to at least some extent) and *neither* (cannot be classified into either of the previous two categories). Table 2 tabulates the distribution of comment categories by treatment. Between *Traditional* and *Pick-the-best*, the rate of the strategy-proofness comment increases by 11.1 percentage points from 10.1% to 21.2% ($p = 0.032$) and the rate of the trade-off comment decreases by 17.2 percentage points from 82.8% to 65.7% ($p = 0.006$). Hence, we observe that the treatment does indeed lead to an increase in the intent to report truthfully by ignoring the priorities at the expense of reporting based on a trade-off between preferences and priorities. Subject comments are highly correlated with their behavior. Whereas subjects with other than a strategy-proofness comment report truthfully in 17.5% of conflict scenarios, subjects with a strategy-proof comment do so in 78.3% of conflict scenarios.¹⁶ This provides further evidence that the treatment effect is driven by an increased fraction

¹⁶One plausible reason for why the latter number is less than 100% is that subjects learn via introspection over time. Indeed, if we focus only on the last 5 rounds, subjects with a strategy-proof comment report truthfully in

Table 2: Frequency of comment categories by treatment

Comment type	Traditional	Pick-the-best
Strategy-proofness	10.1%	21.2%
Trade-off	82.8%	65.7%
Neither	7.1%	13.1%

Figure 3: Probabilities of being assigned to different schools in conflict scenarios (with 95% confidence intervals) by treatment and under truthful reporting



of subjects correctly perceiving that the mechanism is strategy-proof.

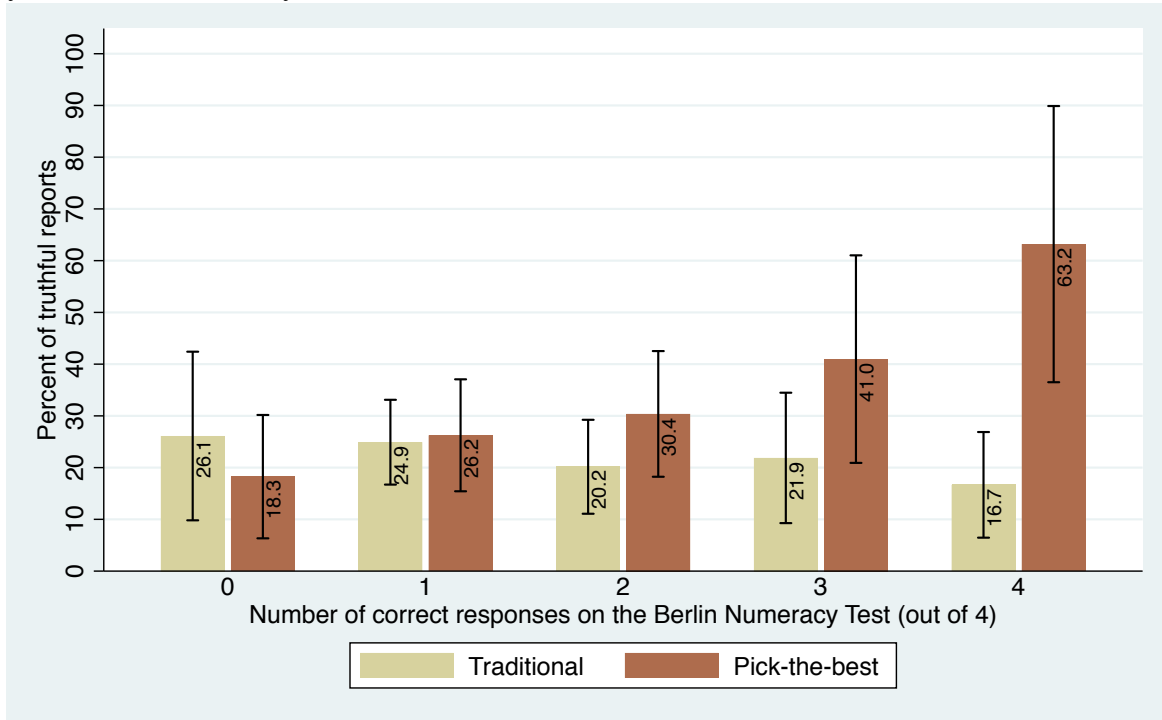
Next, we examine how the treatment-induced shift in the truth-telling rate affects the probability of being assigned to various schools. Using the actual report of a human applicant and the distributions used to generate computerized applicants' submitted rankings and their priorities in various schools (conditional on the given priorities of the human applicant), we compute the probabilities of the human applicant being assigned to each school. The results are presented in Figure 3. The treatment leads to a statistically significant increase in the probability of obtaining *A* from 17.7% to 23.0% ($p = 0.021$) and a statistically significant decrease in the probability of obtaining *B* from 62.1% to 56.5% ($p = 0.007$). There is hardly any difference in the probabilities of obtaining *C* and *D*, respectively, between the two treatments. Hence the pick-the-best description shifts the probability distribution over the four schools in the direction of assigning *A* at the expense of assigning *B*. Examining this shift in more detail, we estimate the probabil-

84.3% of conflict scenarios.

ity of a “costly mistake,” i.e., a non-truthful report that leads to an assignment of a school that is worse than what the subject would get if reporting truthfully. The treatment leads to statistically significant decrease in the probability of a costly mistake from 41.8% to 36.5% ($p = 0.031$). In particular, the probability of being assigned B when A is feasible decreases from 33.9% to 28.4% ($p = 0.005$). These results document that the treatment leads to an expected welfare gain for the human applicants. In addition, Figure 3 also displays the probability distribution of obtaining various schools if reporting truthfully. Subjects in both treatments on average obtain A with a much smaller probability than they would if telling the truth, and obtain each of B , C and D with a (much) higher probability. This illustrates that subjects in both treatments leave money on the table. Across both treatments, in conflict scenarios with a non-truthful report, the average loss in expected payoff relative to reporting truthfully is 23.8%, or €0.95.

5.2 Interaction with numeracy

Figure 4: Average truth-telling rates in the conflict scenarios (with 95% confidence intervals) by the Berlin Numeracy Test score and treatment



The pick-the-best description is designed with the intention to make it easier to perceive the strategy-proofness property. However, the ability to do so requires a certain level of cognitive skills. In this subsection, we investigate how the treatment effect on the average truth-telling rate interacts with a measure of cognitive ability. As introduced in Subsection 4.4, we use the *Berlin Numeracy Test* as a proxy for cognitive ability. We define a subject’s BNT score as the number of correct responses to the four questions on the Berlin Numeracy Test (see Appendix

Table 3: Frequency of reports in the conflict scenarios

Report	Traditional	Pick-the-best
Truth	22.2%	31.8%
<i>BACD</i>	46.5%	33.5%
<i>BCAD</i>	16.6%	17.2%
<i>CBAD</i>	4.2%	4.1%
<i>BCDA</i>	2.0%	3.9%
Other	8.6%	9.4%

E). Appendix C presents summary statistics and distribution of the BNT score by treatment.

Figure 4 plots the truth-telling rate by the BNT score and treatment in the conflict scenarios. Relatively low numbers of subjects in the individual BNT score bins preclude us from conducting bin-by-bin treatment comparisons with sufficient statistical precision. However, we do have sufficient power to evaluate the sign of the interaction between the BNT score and truth-telling separately in each treatment and, hence, also between the BNT score and the treatment effect. We estimate a linear probability model for reporting truthfully with the BNT score, *Pick-the-best* treatment indicator and their interaction as the explanatory variables (along with the constant).¹⁷ While truth-telling is not significantly related to the BNT score in *Traditional* (the estimated slope coefficient of -0.023 , $p = 0.243$), it is highly significantly positively related to the BNT score in *Pick-the-best* (0.094 , $p = 0.002$). Moreover, the estimate of the coefficient on the interaction term is 0.117 ($p = 0.001$).¹⁸ These results provide strong evidence for the hypothesis that perceiving strategy-proofness in *Traditional* is difficult irrespective of cognitive ability, whereas *Pick-the-best* alleviates this difficulty the more the higher one’s cognitive-ability is.

5.3 If not the truth, then what?

We have so far focused on the impact of the treatment on the truth-telling rate, since that is the main objective of the experiment. However, given the truth-telling rates in conflict scenarios of 22.2% in *Traditional* and 31.8% in *Pick-the-best*, one naturally wonders what non-truthful reports subjects actually submit.

Table 3 tabulates the frequencies of the most frequent reports in the conflict scenarios by treatment. By far the two most frequent deviations from truth-telling are *BACD* (about 40% of reports on average across the two treatments) and *BCAD* (16.9%). Two other notable deviations are *CBAD* (4.1%) and *BCDA* (3.0%). Any other deviation has a frequency of less than 1.5% across the two treatments. Given that the applicant has a higher priority in *B* than in *A* in all the conflict scenarios, the evidence presented in Table 3 leads us to conjecture that priorities

¹⁷Standard errors are adjusted for clustering at subject level.

¹⁸We obtain identical qualitative results by using the square or the square root of the BNT score instead of its linear form. This suggests that the qualitative findings are not sensitive to the linear functional form assumption.

tend to drive reporting to a significant extent. In Appendix C, we investigate this relationship in more detail at the level of individual conflict scenarios. This analysis confirms the conjecture. Contrary to the dominant strategy report $ABCD$, subjects often push priority-advantaged schools up the ranking at the expense of priority-disadvantaged schools. This corresponds to the perception of a trade-off subjectively reported by many subjects (Table 2).

We also observe that the treatment-driven gain in truth-telling appears to be offset mostly by a drop in $BACD$, with the relative frequencies of $BCAD$ and $CBAD$ almost unchanged. This drop is statistically significant ($p < 0.001$). However, we do not have any theory for how the treatment should affect the distribution of non-truthful reports. We hence do not feel confident to interpret this result and leave further exploration along these lines to future research.

6 Discussion

The results discussed in the previous section reveal that the pick-the-best mechanism description is indeed effective in increasing the rate of truthful reports in TTC. Moreover, this increase comes from an improved perception of mechanism strategy-proofness at the expense of perceiving existence of a trade-off between preferences and priorities. This is also confirmed by examining the structure of non-truthful reports. A vast majority of them are consistent with submitted rankings being driven by priorities (as opposed to being randomly non-truthful). We speculate that the main culprit for non-truthful reporting is game form misperception, in particular the immediate acceptance perception. This is a generalization of the “district school bias” earlier documented by [Chen and Sönmez \(2006\)](#) and [Pais and Pintér \(2008\)](#). This misperception might be driven by a previous exposure of participants to decentralized non-strategy-proof matching environments involving such feature. For example, sufficiently attractive potential matches in dating or labor markets tend to disappear unless pursued in a timely manner. Hence “shooting for the stars” is risky because, if failing, one returns to a market with fewer and less attractive available options. The perception that there is a trade-off between preferences and feasibility where there is none appears to be more general. For example, [Cason and Plott \(2014\)](#) document that, in the Becker-DeGroot-Marschak mechanism, which is strategically equivalent to SPA, many subjects behave as if participating in a first-price auction (FPA). Again, this could be because many people are much more familiar with FPA than with SPA.

We also observe that the effect of the pick-the-best description on truth-telling interacts positively with cognitive ability. As we argue in Subsection 4.4, solving the tasks on the Berlin Numeracy Test requires contingent thinking. As discussed in Section 2, this ability is also crucial for recognizing strategy-proofness. If interpreting the BNT score as a measure of the ability to think contingently, the positive interaction result can be interpreted as follows. In *Traditional*, even if one can think contingently, recognizing strategy-proofness of the mechanism is, for reasons discussed in Section 1, likely to be difficult. In *Pick-the-best*, complexity is reduced, opening a door for those who can think contingently to spot the strategy-proofness

property.¹⁹

A possible issue with the pick-the-best description is that, unlike a traditional framing, it focuses attention of a player on her own feasible set, potentially raising concerns regarding the overall simultaneous feasibility of the resulting private allocations for all the players. This might hamper a player’s trust in the mechanism (though we do not think the issue of trust in the experimenter is more profound in our experiment relative to other lab experiments). This potential mistrust can be mitigated or eliminated by, for example, employing an independent auditor who would recompute everyone’s allocation using the published mechanism description (with this procedure being pre-announced).

The pick-the-best description is aimed at facilitating the identification of the strategy-proofness property. The description does not include any statements on what constitutes optimal behavior, such as direct action advice (e.g., “You should always rank the institutes in the order of their value to you and ignore priorities.”) or general explanations of consequences of strategy-proofness (e.g., “It is optimal for applicants who only care about getting into the highest-value institute possible to rank the institutes according to their respective values.”). This is mainly for three reasons. First, by restricting ourselves to reformulating the rules of the game, we do not provide potentially wrong or hard to interpret statements. Even though the motivation for the use of strategy-proof mechanisms is based on the assumption that players exhibit “standard” (as compared to “behavioral”) preferences, this might not hold for every player in a field setting. Thus advice is potentially wrong (for at least some players) and/or requires a clear description of the circumstances under which it applies, which we believe is difficult to provide to laymen. Second, current research on providing advice suggests that statements concerning optimal play positively affect rates of truth-telling, in particular if they can be trusted (Guillen and Hakimov, 2018). We believe that trust in the description and execution of a mechanism can be achieved more easily (see above for a suggestion) than trust in a statement advising certain actions to individuals that might more heavily depend on the reputation of the advisor (and is potentially based on hard to explain assumptions of standard preferences and private values). Third, we want to test whether individuals can, in principle, deduce optimality of truth-telling solely from a description of the mechanism. While we believe that additional advice can further increase the rate of truth-telling, we leave this as an open question.

Advice has been shown to be particularly effective in increasing truth-telling if it is given without a description of the mechanism (and any further explanations, see Guillen and Hakimov (2018)). In parallel to this idea, it is possible that at least some of our treatment effect can be attributed to our design choice to relegate the more technical details of the mechanism description to an appendix. While we cannot test this hypothesis with our design, we would like to emphasize that the possibility of separating those elements of the mechanism description that

¹⁹It could also simply be the case that the BNT score is a proxy for attention paid to the experiment. In *Traditional*, it is difficult to recognize strategy-proofness of the mechanism regardless of the level of attention. In *Pick-the-best*, it is easier, but spotting it requires attention.

are sufficient to infer the strategy-proofness property from technical details of the mechanism is a feature of the pick-the-best description. At least for a traditional description, we do not see how such separation could be achieved.

The positive effect of the pick-the-best description on the rate of truthful reporting that we have identified for TTC holds promise for the use of the pick-the-best description for other strategy-proof mechanisms as well. For example, [Gonczarowski et al. \(2023\)](#) document the effectiveness of an analogous mechanism description in increasing truthful reporting in a median voter mechanism. On the other hand, they document that the description is not effective in increasing truth-telling in SPA. This suggests that the effectiveness of the pick-the-best description might depend on the complexity of the original mechanism description and on the degree to which the description simplifies the set of contingencies to consider. More research is necessary to examine the effectiveness of the pick-the-best description for various mechanisms.

7 Conclusion

Cognitive limitations have been suggested as a major driver of preference misreporting in strategy-proof direct-revelation mechanisms. We propose a novel way of describing *any* strategy-proof mechanism with the aim to reduce the cognitive load on players required to identify that the mechanism is strategy-proof. In general, the description looks as follows: “In step 1, based on the reported types of the other players, the mechanism determines which allocations you can feasibly get. Your reported type plays no role in this step. In step 2, the mechanism selects for you a feasible allocation that you prefer the most according to your reported type.” We argue that this description turns the mechanism into a “pick-the-best feasible allocation” problem. It hence becomes easier to evaluate and compare allocations resulting from various type reports.

We experimentally test the impact of such description on truthful type reporting using the top trading cycles mechanism for matching applicants to positions. We focus on “conflict” scenarios that create a tension between applicant preferences and priorities. The aim is to induce an apparent priority-based misreport in case a subject does not perceive the strategy-proofness property of the mechanism. We find that, in the conflict scenarios, our description increases truth-telling by close to one half from 22% to 32%. We also find that, under a traditional description, the truth-telling rate is low and independent of subject numeracy. Under the pick-the-best description, truthful reporting increases with numeracy.

These findings carry several messages that might extend beyond the particular mechanism and laboratory setting that we consider in this paper. First, the proposed description, which comes at no additional cost, holds a potential to improve truth-telling rates in circumstances that are prone to type misreporting. Second, type misreporting in such situations might remain frequent even after the proposed intervention, suggesting we need a better understanding of what drives residual misreporting. Alongside that, we also need to investigate how the proposed intervention compares to and interacts with other possible ways of improving truthful reporting

such as provision of a direct advice on how to report or providing a simulator on which various reporting strategies can be tested. Third, to the extent the numeracy measure reflects cognitive ability and ability to think contingently, the proposed description might create an ability-based inequality in outcomes. On the other hand, to the extent the numeracy measure reflects attention paid to the experiment, we expect the average treatment effect of our description to be larger in field settings as stakes, and hence attention, are much higher there. One way or another, it might be an interesting question for future research to investigate how cognitive ability and attention interact with different ways of improving truthful reporting.

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A Example: Vickrey Auction

In the Vickrey auction, k indivisible objects are allocated to n bidders, with each bidder paying a non-negative price. For $k = 1$, the auction is also referred to as SPA. We denote the set of all k items as K . The set of assignments of the items in K to the seller and bidders is given by $X \equiv \{(y_0, y_1, \dots, y_n) : \bigcup_{i=0}^n y_i = K, y_i \cap y_j = \emptyset \forall i \neq j\}$, with y_0 being the subset of items remaining with the seller and y_i being the subset of items allocated to bidder i . The set of allocations is given by $\mathcal{X} = X \times \{(p_1, \dots, p_n) : p_i \geq 0, i = 1, \dots, n\}$. In this notation, p_i is the price to be paid by bidder i (which can be zero). A bidder i 's type $\theta_i : \mathcal{P}(K) \mapsto \mathbb{R}_+$ is given by her maximum willingness to pay $\theta_i(y_i)$ for each possible subset of items $y_i \subseteq K$ with $\theta_i(\emptyset) = 0$.²⁰ Then $u_{\theta_i}(\cdot)$ is commonly taken to be

$$u_{\theta_i}[(y_0, y_1, \dots, y_n), (p_1, p_2, \dots, p_n)] = \theta_i(y_i) - p_i.$$

This implies that the set $\mathcal{X}_i$ of private allocations of bidder i can be defined as $\mathcal{X}_i \equiv \{(y_i, p_i) : y_i \subseteq K, p_i \geq 0\}$ with

$$h_i((y_0, y_1, \dots, y_n), (p_1, \dots, p_n)) \equiv (y_i, p_i).$$

Then $v_{\theta_i}(\cdot)$ is given by $v_{\theta_i}(y_i, p_i) \equiv \theta_i(y_i) - p_i$. The Vickrey auction asks each bidder i to submit a non-negative bid on every possible subset of items with the restriction that $b_i(\emptyset) = 0$. The vector of bidder i 's bids is denoted by $b_i = (b_i(y_i))_{y_i \in \mathcal{P}(K)}$ and the vector of all bids by $b = (b_1, \dots, b_n)$. The Vickrey auction selects an assignment of items

$$y(b) \in \arg \max_{y \in X} \sum_{i=1}^n b_i(y_i)$$

and sets the payments to

$$p_i(b) = \max_{y \in X} \sum_{j \neq i} b_j(y_j) - \sum_{j \neq i} b_j[y_j(b)].$$

If there are ties, i.e., if $\arg \max_{y \in X} \sum_{i=1}^n b_i(y_i)$ contains more than one element, then the realization λ of the priority ordering determines the assignment of items to bidders (and consequently also the payments). Hence Λ is the set consisting of the $|X|!$ permutations of X . Any $\lambda \in \Lambda$ provides a prioritization of all possible assignments; whichever assignment in the tie comes earliest in the permutation is implemented; that is, the q 'th assignment in λ is prioritized over the r 'th assignment if $q < r$.

Bidder i can obtain any subset of items $y_i \subseteq K$ by bidding arbitrarily high on y_i and zero on any other subset of items. The payment conditional on obtaining y_i is independent of b_i .

²⁰Equivalently, one can think of the type space of player i as $\Theta_i = \mathbb{R}_+^{2^k-1} \times \{0\}$.

Therefore,

$$F_i(b_{-i}, \lambda) = \left\{ (y_i, p_i^{y_i}(b_{-i})) : y_i \subseteq K, p_i^{y_i}(b_{-i}) = \max_{\tilde{y} \in X} \sum_{j \neq i} b_j(\tilde{y}_j) - \max_{\tilde{y} \in X, \tilde{y}_i = y_i} \sum_{j \neq i} b_j(\tilde{y}_j) \right\}.$$

That is, $F_i(b_{-i}, \lambda)$ is a set of price offers, one for each subset of items (including the offer not to obtain any items at a price of zero). For any such set of offers $f_i = \{(y_i, p_i^{y_i}) : y_i \subseteq K\}$, $\tau_{\theta_i}(f_i) = \{(y_i, p_i^{y_i}) : \theta_i(y_i) - p_i^{y_i} = \max_{\tilde{y}_i \subseteq K} \theta_i(\tilde{y}_i) - p_i^{\tilde{y}_i}\}$.

For any set of offers $f_i \in \mathcal{F}_i$, $\tau_{\theta_i}(f_i)$ is the subset of the offers that maximize the net surplus (equal to the realized value minus price paid) of bidder i of type θ_i . f_i is sufficient to determine the payoff of a truthfully reporting player i . This payoff is simply given by $\max_{(y_i, p_i^{y_i}) \in f_i} \theta_i(y_i) - p_i^{y_i}$. However, under typically assumed continuous type spaces, there exists a type θ_i who is indifferent among multiple offers from f_i . For example, in SPA, f_i consists of an offer to buy the object at a price p_i and an offer to not buy the object and to not pay anything. The type $\theta_i = p_i$ is indifferent between these two offers. f_i is therefore not sufficient in general to determine which offer will be chosen on behalf of bidder i . To resolve the tie, the mechanism uses information from (b_{-i}, λ) that is not contained in f_i .

The reframing of the Vickrey Auction could be implemented as follows:

1. “Based on the reported maximum willingnesses to pay (MWPs) of each of the other bidders, the mechanism makes you multiple offers. Each offer specifies a different subset of items and a price at which you can buy this subset. One of the offers is to not buy anything and to not pay anything. *This step does not use your reported MWPs in any way.*
2. The mechanism then accepts exactly one of the offers from step 1 on your behalf. This is an offer for which the difference between your reported MWP and the offered price is maximized over all the offers. In case there are several such offers, the mechanism randomly picks one of them.

B Connection to Partition-Obvious Dominance

Consider partitioning the space of contingencies $\Theta_{-i} \times \Lambda$ of player i into $F_i^{-1}(\mathcal{F}_i)$. In this partition, all contingencies that result in the same feasible set are pooled together into one composite contingency. Given an arbitrary feasible set $f_i \in \mathcal{F}_i$, truthful reporting has a constant payoff $v_{\theta_i}(f_i)$ across all contingencies contained in the composite contingency $F_i^{-1}(f_i)$. Moreover, this payoff is at least as high as the payoff player i could earn from any misreporting strategy under any contingency in $F_i^{-1}(f_i)$. That is, within each composite contingency (but not necessarily across different composite contingencies), truthful reporting obviously dominates (Li, 2017) misreporting. In the terminology of Zhang and Levin (2021), truthful reporting “ $F_i^{-1}(\mathcal{F}_i)$ -partition obviously dominates” misreporting. More formally:

Definition 4. Given a strategy-proof mechanism Γ , let Σ_i be a partition of the set of contingencies $\Theta_{-i} \times \Lambda$ of player i . Truth-telling is said to be **Σ_i -partition obviously dominant** for player i if it is the case for any type $\theta_i \in \Theta_i$ of player i and any $S_i \in \Sigma_i$ that

$$\inf_{(t_{-i}, \lambda) \in S_i} v_{\theta_i}[g_i(\theta_i, t_{-i}, \lambda)] \geq \sup_{(t_i, t_{-i}, \lambda) \in \Theta_i \times S_i} v_{\theta_i}[g_i(t_i, t_{-i}, \lambda)].$$

Proposition 2. Given a strategy-proof mechanism Γ , truth-telling is $F_i^{-1}(\mathcal{F}_i)$ -partition obviously dominant for any player i .

Proof. For any $\theta_i \in \Theta_i$ and for any $f_i \in \mathcal{F}_i$, it is the case for all contingencies (t_{-i}, λ) in the composite contingency $F_i^{-1}(f_i)$ and all possible type reports $t_i \in \Theta_i$ that

$$u_{\theta_i}[g_i(\theta_i, t_{-i}, \lambda)] = V_{\theta_i}(f_i) \geq u_{\theta_i}[g_i(t_i, t_{-i}, \lambda)].$$

Hence

$$\inf_{(t_{-i}, \lambda) \in F_i^{-1}(f_i)} u_{\theta_i}[g_i(\theta_i, t_{-i}, \lambda)] = V_{\theta_i}(f_i) \geq \sup_{(t_i, t_{-i}, \lambda) \in \Theta_i \times F_i^{-1}(f_i)} u_{\theta_i}[g_i(t_i, t_{-i}, \lambda)].$$

□

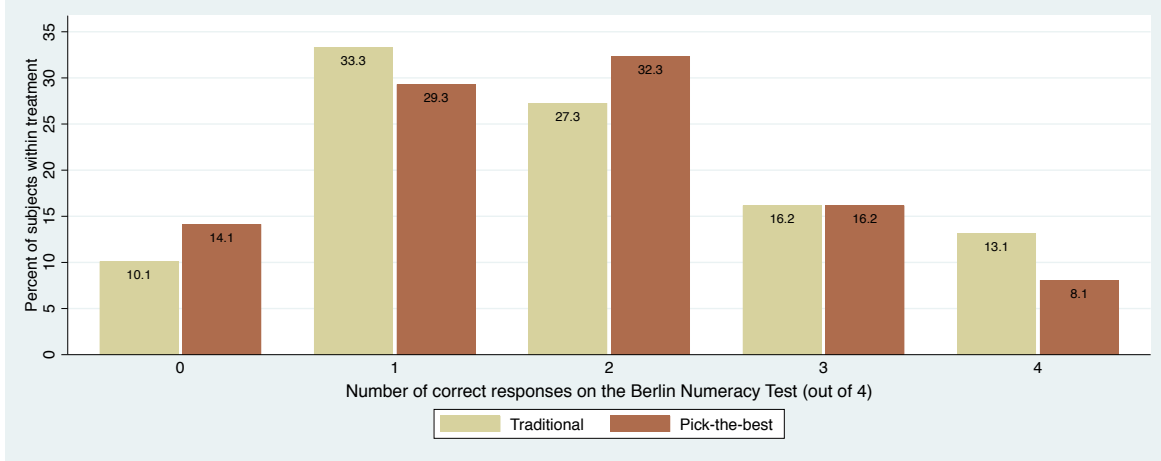
We therefore offer a generally applicable method of constructing a partition of the space of contingencies for the purpose of applying partition dominance. Moreover, the partition constructed this way provides a “general bound” on how coarse such partition can be for any strategy-proof mechanism. The intuition is as follows. A coarser partition requires that at least two composite contingencies $F^{-1}(f_1)$ and $F^{-1}(f_2)$ based on two different feasible sets f_1 and f_2 , respectively, are pooled together. These two feasible sets must differ in at least one private allocation x . Without loss of generality, let $x \in f_1$ but $x \notin f_2$. If there exists a type for which x is strictly preferred over all private allocations in f_2 , then such a type will get something worse than x when facing a contingency from $F^{-1}(f_2)$ and reporting truthfully, but will get x or something even more preferred when facing a contingency from $F^{-1}(f_1)$ and misreporting by re-ranking, relative to the truth, some private allocations that she finds worse than x . This constitutes a violation of obvious dominance on the pooled contingency. This construction is possible if the sets of types are “sufficiently rich” and the number of private allocations is larger than two.

C Supplementary Results

C.1 Distribution of the BNT score in the sample

Figure C.1 displays the distribution of the BNT score by treatment. As expected due to randomization into treatments, the distributions of the BNT score in the two treatments are statistically

Figure C.1: Distribution of subjects over the count of correct responses to the four Berlin Numeracy Test questions



indistinguishable (Pearson’s χ^2 -test $p = 0.638$, Fisher’s exact test $p = 0.649$). The mean score (and its standard deviation) is 1.89 (1.19) in *Traditional* and 1.75 (1.14) in *Pick-the-best*. The median is 2 in both treatments. These distributions are comparable to the ones reported by [Cokely et al. \(2012\)](#) (mean of 1.6, standard deviation of 1.21, median of 2).

C.2 Link between non-truthful reporting and priorities in the conflict scenarios

As mentioned in Subsection 5.3, we hypothesize that priorities tend to drive reporting in the conflict scenarios to a significant extent. That is, contrary to the dominant strategy report *ABCD*, subjects often push priority-advantaged schools up the ranking at the expense of priority-disadvantaged schools. In order to examine this hypothesis, we construct a set of “justifiable” rankings for each scenario. Justifiability is defined as follows: (1) Given a pair of schools, if preferences and priorities over these schools conflict, the two schools might be ranked in either order. (2) If they do not conflict, the two schools are ranked in the order of preferences. Overall, 94.6% of reports are justifiable (94.8% in *Traditional* and 94.5% in *Pick-the-best*). More specifically, 92.7% of non-truthful reports are justifiable (93.3% in *Traditional* and 91.9% in *Pick-the-best*).

The concept of justifiability has predictive power for the location of *C* and *D* in the ranking relative to the other two schools and relative to each other. However, since the priority in *A* is always lower than that in *B* while the payoff is higher, the concept of justifiability is vacuous in relation to the relative position of *A* and *B* in the ranking. To examine the impact of the tension between preferences and priorities along this margin, we turn to the conflict between intensity of the preference for *A* vs. *B* and the intensity of the priority advantage of *B* vs. *A*. To do this, we estimate a linear probability model where the outcome variable is the indicator variable for ranking *A* above *B* and the control variables are the payoff difference between

A and B (between 22 and 38 points) and the priority distance between A and B (between 1 and 3, with a higher number meaning A being more priority-disadvantaged compared to B).²¹ We would expect that the probability of ranking A above B is positively related to the payoff difference and negatively related to the priority difference. We find that each 10 points of the payoff difference increases the probability of ranking A over B by 3.6 percentage points, with the estimate being statistically indistinguishable from 0 ($p = 0.105$). On the other hand, each point of the priority distance *reduces* the probability of ranking A over B by 12.2 percentage points, with the estimate being statistically highly significant ($p < 0.001$). We hence conclude that the probability of ranking A over B decreases with the priority distance between A and B . We obtain qualitatively analogous conclusions if we estimate the model separately for each treatment.

D Instructions

In this section, we present translations of the original instructions that were written in German. These instructions were given to subjects in hardcopy format at the beginning of the experiment. Subjects could refer to them anytime during the experiment. The original instructions are available from the authors upon request.

D.1 Common instructions for the two treatments

GENERAL INFORMATION

Welcome to the AIXperiment laboratory and thank you for your participation in today's experiment. Please do not use the computer for activities other than the experiment. If a pop-up window appears unexpectedly during the experiment (such as a virus scanner), please raise your hand to inform the experimenter. Please do not write anything into these instructions.

You are participating in a decision experiment. In this experiment, you can earn cash money. How much you earn depends on your decisions. In the experiment, your earnings will be counted in points. The number of points you earn during the experiment will be converted into Euros at the end of the experiment and paid to you personally in cash. **Each 20 points correspond to 1 Euro.** If you are to receive less than 4 Euros according to this conversion, we will pay you 4 Euros. We therefore guarantee you a cash payment of at least 4 Euros. Payouts are rounded up to a full 10 cents. Your decisions and entries will be stored anonymously.

If you have any questions after reading these instructions, please raise your hand. The experimenter will then come to you and clarify your question at your seat. In the following, “he” and “his” are used to simplify the presentation. We kindly ask you to interpret this in a gender-neutral way.

²¹We cluster the standard errors at subject level.

During the main part of the experiment, there is no time limit for your input. You can proceed at the speed you wish. The payout starts when all participants in the room are finished. Please remain calmly at your seat until an experimenter calls you to collect your payout.

The rest of these instructions describe the main part of the experiment. When you have finished the main part, you will have an opportunity to earn some extra points. You will then be asked to complete a short demographic questionnaire. At the end of the experiment, you will be shown a summary of your point earnings and the corresponding cash payout and you will be paid out.

MAIN PART OF THE EXPERIMENT

The main part of the experiment consists of **20 independent rounds**. In each round, you will get a job at one of the following four institutes: Circle Institute, Diamond Institute, Square Institute and Triangle Institute. Each institute has exactly one open position. There are a total of four applicants: you and three other applicants whose decisions are made by a computer (hereinafter referred to as computerized applicants). You therefore do not interact with the other participants in the room. At the end of each round, each of the four applicants is assigned to an institute. The way the assignment procedure works in each round is described below.

At the beginning of each round, you will be given a **value** for each of the four institutes. The value of an institute indicates the number of points you will receive in this round if you are employed by that institute at the end of the round. The value usually changes from round to round.

Your cash earnings from the main part are based on three randomly selected rounds. Each round is selected with an equal probability. This random selection is only made after the main part is completed. Therefore, when you make your decisions, you do not know which three rounds will be selected for the payout. You should therefore carefully consider your decision in each round. Also, you should make the decision in one round independently of the decisions in the other rounds.

Until the end of the experiment, you will not receive any information about which institute you were employed at in a particular round. Only at the end of the experiment will you receive an overview of the results showing which institute you were employed at in each round and how many points you received in this round. The results overview will also show which three rounds were selected for your cash payout.

APPLICATION PROCEDURE

At the beginning of the round, each of the four institutes ranks the four applicants according to their attractiveness. This happens by means of a **list of priorities**. Such a list of priorities consists of the numbers 1, 2, 3 and 4, with 1 being the highest and 4 the lowest priority. From each institute, each of the four applicants is assigned one of these numbers, whereby no two applicants receive the same number. The applicant with priority 1 is preferred by the institute to the applicant with priority 2, and so on. Different institutes typically rank the applicants differently.

At the beginning of the round, you will be informed of the priorities you have at each of the four institutes. The three remaining priorities within each institute (that is, the priorities that are not equal to your priority at that institute) are randomly distributed among the three computerized applicants. For example, if your priority at an institute is 2, the remaining priorities within the institute, 1, 3 and 4, will be randomly distributed among the three computerized applicants. Each of the computerized applicants receives each of the three remaining priorities with an equal probability. You will not be told the priorities of the other applicants. **For example**, the lists of priorities could be as follows (with the three computerised applicants labelled as CA1-CA3):

Applicant	Priorities of applicants in the institutes			
	Triangle	Circle	Square	Diamond
You	2	1	2	4
CA1	3	2	4	1
CA2	4	3	1	3
CA3	1	4	3	2

You will only find out your priorities (in bold above) in the institutes.

At the beginning of the round, you will also be informed about your respective point values for the four institutes. After you have been informed of your values for the institutes and your priority at each institute, you provide a **ranking of the four institutes**. You do this by giving each of the four institutes a rank, from highest rank (rank 1) to lowest rank (rank 4). Each of the three computerized applicants will also give a ranking. The rankings of the computerized applicants are determined randomly, with each of the possible 24 rankings of an applicant being equally likely.

Note that a ranking of the institutes is always referred to as a ranking (with ranks). A ranking of applicants is always referred to as a list of priorities (with priorities).

The lists of priorities and the submitted rankings of the computerized applicants are determined **anew in each round**. Page 5 describes in detail the procedure for determining which institute will employ you given the submitted rankings and priority lists.

Below is an example of the interface used to enter your ranking in each round.

Round 3

Enter your ranking in the input fields!

Institute	Value	Your priority in the institute	Your rank for the institute
Square Institute	3	2	Enter 1, 2, 3 or 4!
Circle Institute	66	3	Enter 1, 2, 3 or 4!
Diamond Institute	32	1	Enter 1, 2, 3 or 4!
Triangle Institute	92	3	Enter 1, 2, 3 or 4!

[Continue](#)

In the first column, you see the names of the four institutes. The line of an institute contains in the second column the point value you receive for the institute (that is, the number of points you receive when you are assigned to this institute) and in the third column your priority at this institute. You enter your rank for the institute in the box in the fourth column.

You can only enter one of the numbers 1, 2, 3 or 4 in each box. Also, you can only enter each of the four numbers once and no box can be left empty.

On the next page, we describe the utilized assignment procedure.

D.2 Instructions specific to *Traditional*

ASSIGNMENT PROCEDURE

An assignment procedure consisting of one or more iterations is used.

In **iteration 1**, it is assumed that each candidate points to the institute he gave the highest rank to. It is further assumed that each institute points to the applicant with the highest priority. The procedure now finds all the resulting cycles. A cycle consists of a sequence of applicants and institutes in which the first applicant in the sequence points to an institute, which again points to an applicant, which again points to an institute, and so on until the last institute in the sequence points to the first applicant. Such a cycle contains an equal number of applicants and institutes. There can be one or more cycles. If you are included in a cycle, you will be employed by the institute you are pointing to.

If you are not included in any cycle, the procedure is repeated in **iteration 2**, but only with the institutes and applicants that were not included in **any** of the cycles in iteration 1 (which means you are included in iteration 2). We call these the remaining institutes and the remaining applicants. If there are at least

two remaining applicants (and then also two remaining institutes), it is now assumed that each remaining applicant points to the remaining institute he gave the highest rank to among all the remaining institutes. It is also assumed that each remaining institute points to the remaining applicant who has the highest priority among all the remaining applicants. If you are the only remaining applicant, you point to the only remaining institute and it points to you. The procedure finds all the resulting cycles. There can be one or more cycles. If you are included in a cycle, you will be employed by the remaining institute you are pointing to.

If you were not previously included in any cycle, further iterations are carried out in the same way as iterations 1 and 2 until you are assigned. The procedure requires a maximum of 4 iterations.

On the following pages, we illustrate the assignment procedure using examples. **These are for illustrative purposes only** and do not give any indication of the ranking you should give in the experiment or what the ranking of the computerized applicants or the priorities of the institutes will be.

HOW THE ASSIGNMENT PROCEDURE WORKS: EXAMPLE 1

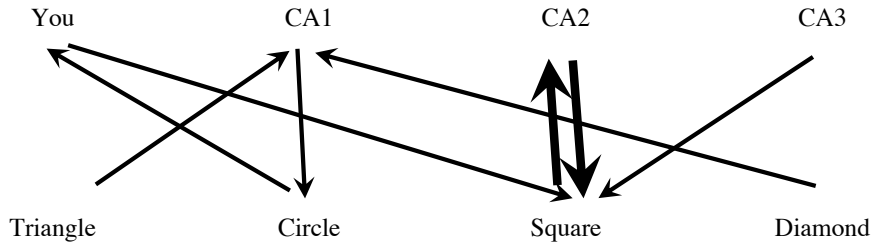
Assume that **you** have submitted a ranking in which you gave Square Institute rank 1 (the highest rank), Diamond Institute rank 2 (the second highest rank), Circle Institute rank 3 (the third highest rank) and Triangle Institute rank 4 (the lowest rank). Further, assume that the three **computerized applicants** we designate as CA1, CA2 and CA3 have given **rankings** according to the following table (which also includes your ranking):

	Applicants' rankings			
Institute	You	CA1	CA2	CA3
Triangle	4	4	2	2
Circle	3	1	3	4
Square	1	3	1	1
Diamond	2	2	4	3

Assume that the **lists of priorities** of the institutes are as follows:

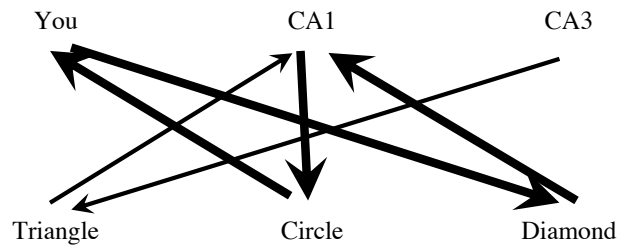
	Priorities of applicants in the institutes			
Applicant	Triangle	Circle	Square	Diamond
You	2	1	2	3
CA1	1	2	3	1
CA2	4	3	1	4
CA3	3	4	4	2

In **iteration 1**, it is assumed that each candidate points to the institute he gave the rank 1 to. Each institute points to the applicant who has the highest priority 1 at the institute. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. You are not part of the cycle. The procedure therefore continues with iteration 2.

In **iteration 2**, the institutes and applicants who were not included in the cycle of iteration 1 remain. Each remaining applicant points to the remaining institute which he gave a higher rank to than the other remaining institutes. Furthermore, each remaining institute points to the remaining applicant who has a higher priority in that institute than the other remaining applicants. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. You are part of the cycle and are therefore employed at the institute you point to, that is, Diamond Institute.

HOW THE ASSIGNMENT PROCEDURE WORKS: EXAMPLE 2

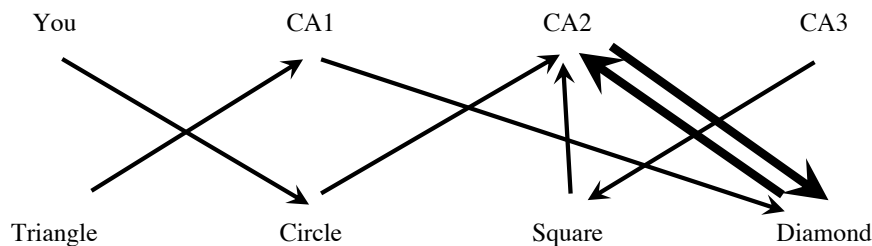
Suppose that the following **rankings** have been submitted:

	Applicants' rankings			
Institute	You	CA1	CA2	CA3
Triangle	2	2	4	3
Circle	1	3	3	2
Square	4	4	2	1
Diamond	3	1	1	4

Assume that the **lists of priorities** of the institutes are as follows:

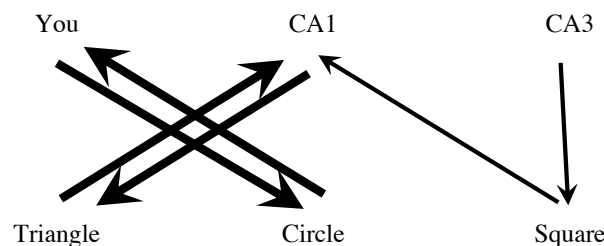
	Priorities of applicants in the institutes			
Applicant	Triangle	Circle	Square	Diamond
You	3	2	3	2
CA1	1	4	2	3
CA2	2	1	1	1
CA3	4	3	4	4

In **iteration 1**, it is assumed that each candidate points to the institute which he gave rank 1 to. Each institute points to the applicant who has the highest priority in the institute. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. You are not part of the cycle. The procedure therefore continues with iteration 2.

In **iteration 2**, the institutes and applicants who were not included in the cycle of iteration 1 remain. Each remaining applicant points to the remaining institute which he gave a higher rank to than the other remaining institutes. Furthermore, each remaining institute points to the remaining applicant who has a higher priority in that institute than the other remaining applicants. The procedure finds all cycles.



There are two cycles, which are marked by the highlighted arrows. You are part of a cycle and are therefore employed at the institute you are pointing to, that is, Circle Institute.

D.3 Instructions specific to *Pick-the-best*

ASSIGNMENT PROCEDURE

An assignment procedure is used which consists of two steps.

In **step 1**, the procedure uses the rankings provided by the three computerised applicants and the lists of priorities of the four institutes to determine which institutes are available to you (**the available institutes** to you) and which are not (**the unavailable institutes** to you). There will always be at least one available institute.

Note that step 1 does not use your ranking. Whether or not an institute is available to you does not depend on your ranking.

In **step 2**, the procedure uses the result of step 1 and your ranking and assigns you to the available institute with the highest rank. Specifically, the procedure first considers the institute which you gave the highest rank (rank 1) to. If this institute is available to you, you will be employed there and the procedure is finished. If this institute is not available to you, the procedure next considers the institute which you gave the second highest rank (rank 2) to. If this institute is available to you, you will be employed there and the procedure is finished. If both this institute and the institute you gave rank 1 to are unavailable to you, the procedure next considers the institute you gave the third highest rank (rank 3) to. If this institute is available to you, you will be employed there and the procedure is finished. If the institutes you gave the ranks 1, 2 or 3 to are unavailable to you, the procedure next considers the institute you gave the fourth highest rank (rank 4) to. Since at least one institute is available to you, this institute must be available to you in this case and you will be employed there.

Please note that, given the result of step 1, your employment depends solely on your ranking and not on your priorities at the four institutes.

On the following pages, we illustrate step 2 of the assignment procedure using examples. **These are for illustrative purposes only** and do not give any indication of what ranking you should submit in the experiment or what the lists of available institutes in the experiment look like. Details on the exact functioning of step 1 are given in the Appendix.

HOW THE ASSIGNMENT PROCEDURE WORKS: EXAMPLE 1

The lists of priorities of the institutes and the rankings of the computerised applicants determined that the following three institutes are available to you: Triangle, Circle and Diamond. Details of how the institutes that are available to you have been determined are described in Example 1 in the Appendix.

Suppose you gave a ranking as in the table below. In the table, the institutes that are available to you are underlined.

Rank	1	2	3	4
Institute	Square	<u>Diamond</u>	<u>Circle</u>	<u>Triangle</u>

The procedure now first considers Square Institute because you gave it the highest rank. Since this institute is not available to you, the procedure next considers the institute you gave rank 2 to, that is, Diamond Institute. Since this institute is available to you, you will be employed at **Diamond Institute**.

HOW THE ASSIGNMENT PROCEDURE WORKS: EXAMPLE 2

The lists of priorities of the institutes and the rankings of the computerised applicants determined that the following two institutes are available to you: Circle and Square. Details of how the institutes that are available to you have been determined are described in Example 2 in the Appendix.

Suppose you gave a ranking as in the table below. In the table, the institutes that are available to you are underlined.

Rank	1	2	3	4
Institute	<u>Circle</u>	Triangle	Diamond	<u>Square</u>

The procedure now first considers Circle Institute, since you gave it the highest rank. Since this institute is available to you, you will be employed at **Circle Institute**.

D.4 Appendix for *Pick-the-best*

DETERMINATION OF AVAILABILITY

The assignment procedure checks separately for each of the four institutes whether it is available to you or not. To do this, the procedure uses the rankings given by the three computerized applicants (**but not the ranking you gave**) and the lists of priorities of the institutes. To check whether a specific institute is available to you, the procedure goes through one or more iterations as follows:

In **iteration 1**, it is assumed that you point to this specific institute. Each computerized applicant points to the institute which he gave the highest rank to. It is further assumed that each institute points to the candidate with the highest priority. The procedure now finds all the resulting cycles. A cycle consists of a sequence of applicants and institutes in which the first applicant in the sequence points to an institute, which again points to an applicant, which again points to an institute, and so on until the last institute in the sequence points to the first applicant. Such a cycle contains an equal number of applicants and institutes. There can be one or more cycles. If you are included in such a cycle, the specific institute is **available** to you. If you are not included in a cycle, but the specific institute is included in a cycle, then the specific institute is **not available** to you.

If neither you nor the specific institute is included in a cycle of iteration 1, the procedure is repeated in **iteration 2**, but only with those institutes and applicants that were not included in **any** of the cycles of iteration 1 (which means that you are part of it). We call these the remaining institutes and the remaining applicants. It is further assumed that you point to the specific institute and each remaining computerized applicant points to the remaining institute which he gave the highest rank to among the remaining institutes. It is also assumed that each remaining institute points to the remaining applicant who has the highest priority among the remaining applicants. The procedure finds all the resulting cycles. There may be one or more cycles. If you are included in such a cycle, then the specific institute is **available** to you. If you are not included in any cycle, but the specific institute is included in a cycle, then the specific institute is **not available** to you.

If neither you, nor the specific institute were included in a cycle in iteration 1 or iteration 2, further iterations are carried out analogous to iterations 1 and 2 until it is determined whether the specific institute is available to you or not. The procedure requires a maximum of 4 iterations.

On the following pages, we illustrate the determination of the available institutes by means of examples that match the examples in the Instructions. These examples are based on rankings given by the computerised applicants and on lists of priorities of the institutes. **Their choice serves exclusively to illustrate**

the assignment procedure. The concrete choice of rankings and priorities does not give any indication of the rankings of the computerized applicants or the priorities of the institutes in the experiment.

EXAMPLE 1

Suppose that the three **computerised applicants** CA1, CA2 and CA3 have submitted the following rankings:

	Computerized applicants' rankings		
Institute	CA1	CA2	CA3
Triangle	4	2	2
Circle	1	3	4
Square	3	1	1
Diamond	2	4	3

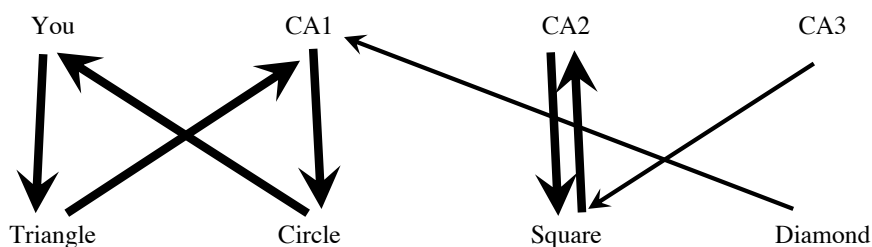
Assume that the **lists of priorities** of the institutes are as follows:

	Priorities of applicants in the institutes			
Applicant	Triangle	Circle	Square	Diamond
You	2	1	2	3
CA1	1	2	3	1
CA2	4	3	1	4
CA3	3	4	4	2

In this example, **Triangle Institute**, **Circle Institute** and **Diamond Institute** are available to you, whereas **Square Institute** is not available to you. The following explains why this is so.

Determining whether Triangle Institute is available to you:

In **iteration 1**, it is assumed that you point to Triangle Institute. Each computerized applicant points to the institute which he gave rank 1 to. Each institute points to the applicant who has the highest priority 1 with the institute. The procedure finds all cycles.



There are two cycles, which are marked by the highlighted arrows. You are part of a cycle. Triangle Institute is therefore available to you.

Determining whether Circle Institute is available to you:

In **iteration 1**, it is assumed that you point to Circle Institute. Each computerized applicant points to the institute to which he gave rank 1 to. Each institute points to the applicant who has the highest priority 1 in the institute. The procedure finds all cycles.

Suppose that the three **computerised applicants** CA1, CA2 and CA3 have submitted the following rankings:

	Computerized applicants' rankings		
Institute	CA1	CA2	CA3
Triangle	2	4	3
Circle	3	3	2
Square	4	2	1
Diamond	1	1	4

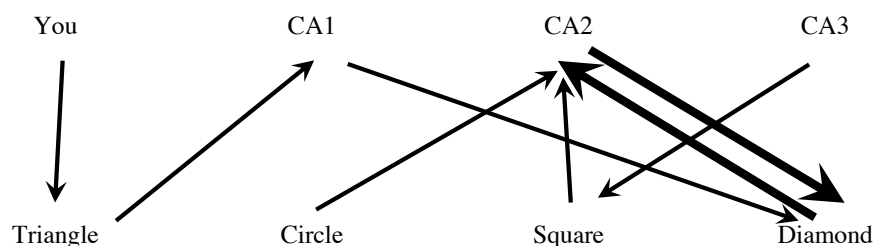
Suppose that the **lists of priorities** of the institutes are as follows:

	Priorities of applicants in the institutes			
Applicant	Triangle	Circle	Square	Diamond
You	3	2	3	2
CA1	1	4	2	3
CA2	2	1	1	1
CA3	4	3	4	4

In this example, **Circle Institute and Square Institute are available to you**, whereas **Triangle Institute and Diamond Institute are not available to you**. The following explains why this is so.

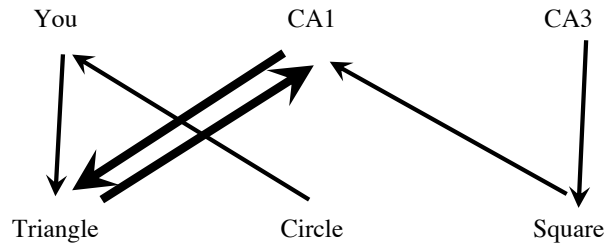
Determining whether Triangle Institute is available to you:

In **iteration 1**, it is assumed that you point to Triangle Institute. Each computerized applicant points to the institute which he gave rank 1 to. Each institute points to the applicant who has the highest priority 1 in the institute. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. Neither you nor Triangle Institute are part of the cycle. The procedure therefore continues with iteration 2.

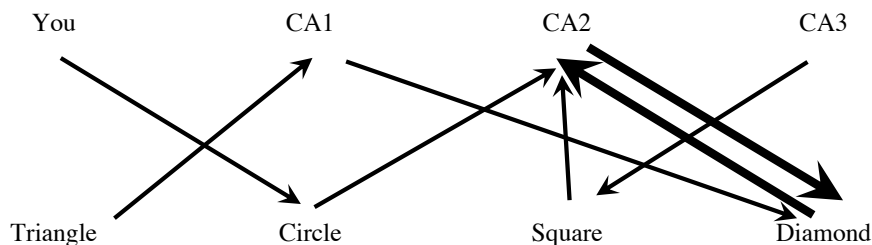
In **iteration 2**, the institutes and applicants who were not included in the cycle of iteration 1 remain. You point to Triangle Institute and each remaining computerised applicant points to the remaining institute which he gave a higher rank to than the other remaining institutes. Furthermore, each remaining institute points to the remaining applicant who has a higher priority than the other remaining applicants. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. You are not part of the cycle, whereas Triangle Institute is part of the cycle. Triangle Institute is therefore not available to you.

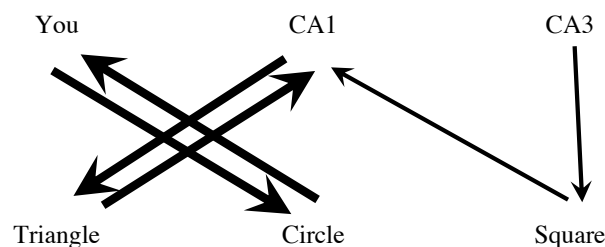
Determining whether Circle Institute is available to you:

In **iteration 1** it is assumed that you point to Circle Institute. Each computerized applicant points to the institute to which he has given rank 1 to. Each institute points to the applicant who has the highest priority 1 in the institute. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. Neither you nor the Circle Institute are part of the cycle. The procedure therefore continues with iteration 2.

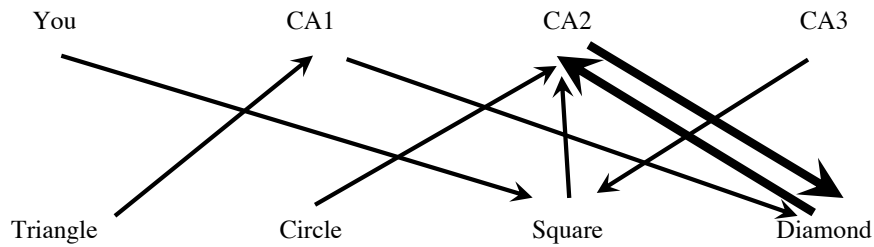
In **iteration 2**, the institutes and applicants who were not included in the cycle of iteration 1 remain. You point to Circle Institute and each remaining computerized applicant points to the remaining institute which he gave a higher rank to than the other remaining institutes. Furthermore, each remaining institute points to the remaining applicant who has a higher priority in that institute than the other remaining applicants. The procedure finds all cycles.



There are two cycles, which are marked by the highlighted arrows. You are part of a cycle. Circle Institute is therefore available to you.

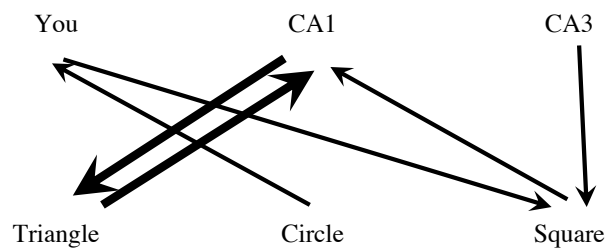
Determining whether Square Institute is available to you:

In **iteration 1**, it is assumed that you point to Square Institute. Each computerized applicant points to the institute which he gave rank 1 to. Each institute points to the applicant who has the highest priority 1 in the institute. The procedure finds all cycles.



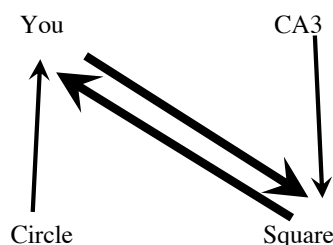
There is a cycle, which is marked by the highlighted arrows. Neither you nor Square Institute are part of the cycle. The procedure therefore continues with iteration 2.

In **iteration 2**, the institutes and applicants who were not included in the cycle of iteration 1 remain. You point to Square Institute and each remaining computerized applicant points to the remaining institute which he gave a higher rank to than the other remaining institutes. Furthermore, each remaining institute points to the remaining applicant who has a higher priority in that institute than the other remaining applicants. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. Neither you nor Square Institute are part of the cycle. The procedure therefore continues with iteration 3.

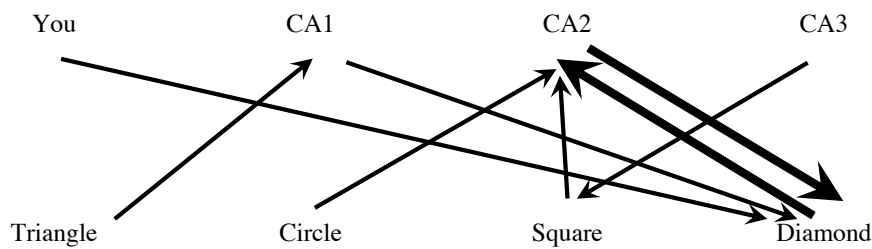
In **iteration 3**, the institutes and applicants who were not included in the cycle of iteration 1 or iteration 2 remain. You point to Square Institute and the remaining computerized applicant points to the remaining institute which he gave a higher rank to than the other one. Furthermore, each remaining institute points to the remaining applicant who has the higher priority. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. You are part of the cycle. Square Institute is therefore available to you.

Determining whether the Raute-Institut is available to you:

In **iteration 1**, it is assumed that you point to Diamond Institute. Each computerized applicant points to the institute which he gave rank 1 to. Each institute points to the applicant who has the highest priority 1 in the institute. The procedure finds all cycles.



There is a cycle, which is marked by the highlighted arrows. You are not part of the cycle, whereas Diamond Institute is part of the cycle. Diamond Institute is therefore not available to you.

E Berlin Numeracy Test questions

Here we present the four multiple-choice questions and the offered answers from the *Berlin Numeracy Test*. The wording in English is taken directly from [Cokely et al. \(2012\)](#). For use in the experiment, we translate these questions and answers into German. The translation is available from the authors upon request. Here we add boldface emphasis on the correct responses.

Instructions: Please answer the questions below. Do not use a calculator but feel free to use the space available for notes (i.e., scratch paper).

1. Imagine we are throwing a five-sided die 50 times. On average, out of these 50 throws how many times would this five-sided die show an odd number (1, 3 or 5)

- a) 5 out of 50 throws
- b) 25 out of 50 throws
- c) **30 out of 50 throws**
- d) None of the above

2. Out of 1,000 people in a small town 500 are members of a choir. Out of these 500 members in the choir 100 are men. Out of the 500 inhabitants that are not in the choir 300 are men. What is the probability that a randomly drawn man is a member of the choir? Please indicate the probability in percent

- a) 10%
- b) **25%**
- c) 40%

d) None of the above

3. Imagine we are throwing a loaded die (6 sides). The probability that the die shows a 6 is twice as high as the probability of each of the other numbers. On average, out of these 70 throws, about how many times would the die show the number 6?

a) **20 out of 70 throws**

b) 23 out of 70 throws

c) 35 out of 70 throws

d) None of the above

4. In a forest 20% of mushrooms are red, 50% brown and 30% white. A red mushroom is poisonous with a probability of 20%. A mushroom that is not red is poisonous with a probability of 5%. What is the probability that a poisonous mushroom in the forest is red?

a) 4%

b) 20%

c) **50%**

d) None of the above